

Conditional Survival and Nomogram for Elderly Non-Metastatic Colon Cancer Patients Following Colectomy

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Abstract

Background

This study aimed to evaluate the conditional survival (CS) of elderly patients with non-metastatic colon cancer who underwent colectomy and build conditional nomograms that can accommodate varying survival periods and estimate survival rates.

Methods

Data from 9302 patients between 2004 and 2017 were obtained from the Surveillance, Epidemiology, and End Results database. CS was used to assess overall survival and colon-specific survival rates in patients who survived beyond a certain time period. Cox regression was used to select factors for nomogram development, and performance was evaluated using area under the receiver operating characteristic curve (AUC), calibration plot, and decision curve analysis (DCA).

Results

The 5-year conditional overall survival rates initially increased slightly but then decreased over time. The rates at different time points after diagnosis (baseline and 1-5 years) were 62.5, 63.2, 62.8, 62.1, 61.6, and 59.8%. In contrast, 5-year conditional colon-specific survival rates consistently improved over the same period. These rates were 78.1, 80.9, 84.2, 86.9, 89.3, and 90.9%, respectively. Nomograms were developed for baseline measurements and for patients who survived 1, 3, and 5 years. The performance of these nomograms, assessed using AUC, calibration curves, and DCA, indicated good predictive capabilities.

Conclusion

CS provides valuable information on the medium- and long-term survival probabilities of elderly patients with non-metastatic colon cancer who underwent colectomy. The developed conditional nomograms allowed for the estimation of survival probabilities across different timeframes, facilitating a comprehensive understanding of prognosis and collaborative decision making.

eLife assessment

This **useful** work provides a risk-prediction tool, in the form of a nomogram, for practitioners and elderly patients with non-metastatic colon cancer using data from the SEER registry. The unique contribution of this work is the focus on conditional survival. However, the underlying statistical approach is suboptimal and therefore **incomplete**, which substantially lessens the potential impact of this work. The analysis could use a more rigorous consideration of competing risks.

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1. Introduction

Colon cancer is the third most common and deadly form of cancer worldwide, accounting for 10% of all cancer cases¹. It mainly affects the elderly population, with most patients aged > 60 years and a significant number aged > 75². As the global population ages, a significant increase in the prevalence of colon cancer is expected among older individuals. In the United States alone, the elderly population (65 years and older) is expected to increase by 20% by 2030, leading to a parallel surge in colon cancer cases³. Additionally, the number of adults aged ≥ 85 years is expected to triple to 19 million by 2060, further contributing to the rise in colon cancer cases. While there is no standardized definition of “elderly” in relation to colon cancer, it generally includes individuals aged 65-75 or older⁴⁻⁶.

Surgery is the preferred treatment for stage I–III colon cancer. Advances in medical standards have made radical surgery more feasible in elderly patients. Studies have shown an increase in the rate of radical surgery among colon cancer patients aged ≥ 75 years who were diagnosed with stage I or II⁷. Surgical resection has been found to improve the prognosis of elderly colon cancer⁸⁻⁹. Improved perioperative management and comprehensive geriatric assessments have contributed to better tolerance to surgical treatment and improved quality of life in the elderly¹⁰. These factors have led to prolonged survival of elderly patients undergoing surgery.

Conditional survival (CS) is a statistical measure that estimates the likelihood of survival for a specific duration after diagnosis or treatment, considering the patient’s current survival time¹¹. This approach provides a more personalized prognosis that evolves over time, making it more relevant and meaningful to patients than the traditional survival analysis. It allows physicians to tailor checkup schedules and content based on a patient’s unique survival pattern, enabling personalized adjustments. Previous studies on conditional survival in colorectal cancer have been conducted¹²⁻¹⁴, but most have included a diverse age range, and few have focused exclusively on the older population. A nomogram, which is a predictive tool for prognosis, has emerged as valuable in assessing the chances of survival in elderly colon cancer patients¹⁵. However, the

nomogram overlooks the crucial aspect of incorporating patient survival duration. This study aimed to investigate conditional survival rates in elderly non-metastatic colon cancer patients who underwent colectomy, while developing nomograms to assess their prognosis.

2. Materials and methods

Data collection

This study used data from the November 2020 release of the Surveillance, Epidemiology, and End Results (SEER) database obtained specifically from SEER*Stat Version 8.4.2. The study adhered to the guidelines of the 1964 Declaration of Helsinki and was conducted according to the STROCSS criteria. As per our institute's Ethics Committee, this study did not require approval from an institutional review board because it used de-identified public database information. The study included a subset of elderly colon cancer patients who underwent colectomy and met the following criteria: (1) aged 70–89 years, (2) pathologically diagnosed with stage I–III colon cancer, (3) had undergone colectomy, and (4) had colon cancer as the only primary cancer. Patients who received neoadjuvant therapy were excluded from the analysis because of their potential effects on downstaging and prognostic factors. Patients without clinicopathological data or survival information were excluded from this study. A diagram outlining the screening process is presented in **Supplementary Figure 1** [\[1\]](#).

Statistical analyses

In this study, the time from diagnosis to death from any cause was referred to as overall survival (OS), while the time from diagnosis to death, specifically from colon cancer, was referred to as colon-specific survival (CSS). A multivariate Cox regression model was constructed to assess the correlation between clinicopathological characteristics and survival outcomes. Survival curves were generated using the Kaplan-Meier method, and survival disparities were assessed using the log-rank test. CS represents the likelihood of survival for an additional specified time period (y years) after survival for a certain duration (x years). This measure was derived from Kaplan-Meier survival data and was mathematically expressed as $CS(x|y) = S(x+y)/S(x)$, where $S(x)$ represents the OS or CSS rate at x years, as estimated the Kaplan-Meier method¹⁶ [\[2\]](#). This study employed COS (x) and CCSS (x) as alternatives to $S(x)$ to determine the number of survivors in year x . The standardized differences (d) method was used to compute variations in CS across the subgroups¹⁷ [\[3\]](#). When $|d|$ was less than 0.1, it indicated negligible difference within each group; when it ranged from 0.1 to less than 0.3, it signified a minor difference; when it fell between 0.3 and less than 0.5, it suggested a moderate difference; and when $|d|$ was 0.5 or greater, it indicated a notable difference.

The study participants were randomly allocated to either the training or validation cohort with a distribution ratio of 7:3. Univariate Cox regression analysis was performed to identify potential variables for inclusion in multivariate analysis, with a requirement for significance of $p < 0.05$. Subsequently, variables demonstrating prognostic significance in multivariate Cox regression analysis were used to construct nomograms. To evaluate the discriminative ability of the nomogram, the area under the receiver operating characteristic (ROC) curve (AUC) was calculated. The calibration assessment was conducted using the Brier score and a calibration plot, employing the bootstrapping approach with 1000 resampling bootstraps. After the nomogram was subjected to decision curve analysis (DCA), its clinical efficacy was assessed.

R version 4.1.1 was employed for statistical analysis. Categorical variables are presented as frequencies and percentages, and disparities among various groups were examined using the chi-squared test. Continuous variables were expressed as means with standard deviations and compared using the t-test or rank-sum test based on data distribution. Statistical significance was determined using a p -value less than 0.05.

3. Results

Clinicopathological characteristics

This study encompassed 9302 cases from 2004 to 2017 based on specific criteria. As shown in **Table 1**, the training set consisted of 6511 patients, whereas the validation set comprised 2791 patients. The patients had an average age of 78.6 years with a standard deviation of 5.42. The majority of these patients were female (57.4%), white (81.0%), diagnosed after 2010 (55.2%), and had tumors located in the proximal colon (73.1%). Grade II patients constituted the largest proportion (70.2%), followed by grade III (19.7%), grade I (6.64%), and grade IV (3.48%) patients. Colon adenocarcinoma was the most common cancer type (89.7%). Tumors < 5 cm accounted for 58.0% of the cases, and the majority of patients were married (51.6%) and had negative or normal carcino-embryonic antigen (CEA) test results (62.6%). Tumor stages were classified as T1 (7.99%), T2 (16.7%), T3 (61.2%), and T4 (14.1%). Nodal stages were classified as N0 (63.4%), N1 (24.6%), and N2 (12.0%). A small proportion of the patients received radiotherapy (0.8%) or chemotherapy (22.5%). The training and validation cohorts exhibited no significant differences in any of the variables.

COS and CCSS

Figure 1 shows the conditional survival curves. The analysis demonstrated a consistent pattern of improvement in the actuarial OS and CSS of individuals as the duration of survival increased annually. Meanwhile, as shown in **Table 2** and **Figure 1**, the COS5 rates initially saw a marginal uptick, followed by a subsequent decline. Specifically, the rates at baseline, one year, two years, three years, four years, and five years post-diagnosis were 62.5%, 63.2%, 62.8%, 62.1%, 61.6%, and 59.8%, respectively. However, the CCSS5 rates exhibited an upward trajectory over the five-year period, with the initial and subsequent conditional survival rates at baseline and years 1 through 5 being 78.1%, 80.9%, 84.2%, 86.9%, 89.3%, and 90.9%, respectively (**Table 3**).

Subsequently, detailed subgroup analysis was conducted to assess the impact of various factors on the rates of COS5 and CCSS5. The expression rates of COS5 and CCSS5 varied across these subgroups, displaying distinct patterns over time. For COS5, there were persistent negligible or minor differences in survival between some subgroups ($|d| < 0.1$: ≥ 2010 vs. < 2010 , Distal vs. Proximal, Grade II vs. Grade I, and T2 vs. T1; $0.1 < |d| < 0.3$: Widowed vs. Married and Separated/Divorced vs. Married). The differences in COS5 levels between some subgroups gradually appeared over time ($|d| < 0.1 \rightarrow |d| > 0.1$: Others vs. White and Chemotherapy Yes vs. None/Unknown). The disparity in COS5 rates among some subgroups slowly diminished over time ($|d| > 0.1 \rightarrow |d| < 0.1$, $|d| > 0.3 \rightarrow |d| < 0.3$, or $|d| > 0.5 \rightarrow |d| < 0.5$; black vs. White, Others vs. proximal, grade III vs. Grade I, Grade IV vs. Grade I, Single vs. Married, Positive/Elevated vs. Negative/Normal, T3 vs. T1, T4 vs. T1, N1 vs. N0, and N2 vs. N0). Over time, the variance in the COS5 rate fluctuated irregularly across other subgroups. Correspondingly, for CCSS5, there were persistent negligible differences in survival between some subgroups ($|d| < 0.1$: others vs. white, ≥ 2010 vs. < 2010 , mucinous adenocarcinoma vs. adenocarcinoma, owed vs. Married and T2 vs. T1). The variance in CCSS5 among the subgroups became increasingly evident as time progressed ($|d| < 0.1 \rightarrow |d| > 0.1$: Distal vs. Proximal). Over time, there was a gradual reduction in the differences in CCSS5 rates within certain subgroups ($|d| > 0.1 \rightarrow |d| < 0.1$, $|d| > 0.3 \rightarrow |d| < 0.3$, or $|d| > 0.5 \rightarrow |d| < 0.5$: black vs. White, Grade II vs. Grade I, Grade IV vs. Grade I, ≥ 5 cm vs. < 5 cm, single vs. Married, Positive/Elevated vs. Negative/Normal, T3 vs. T1, T4 vs. T1, N1 vs. N0, and N2 vs. N0). The differences in the CCSS5 rate among the other subgroups fluctuated irregularly over time.

	Total (N=9302)	Train (N=6511)	Test (N=2791)	p
Age	78.6 (5.42)	78.6 (5.44)	78.6 (5.40)	0.894
Race				0.360
White	7533 (81.0%)	5293 (81.3%)	2240 (80.3%)	
Black	532 (5.72%)	359 (5.51%)	173 (6.20%)	
Others	1237 (13.3%)	859 (13.2%)	378 (13.5%)	
Sex:				0.744
Male	3964 (42.6%)	2767 (42.5%)	1197 (42.9%)	
Female	5338 (57.4%)	3744 (57.5%)	1594 (57.1%)	
Year of Diagnosis:				0.643
< 2010	4165 (44.8%)	2926 (44.9%)	1239 (44.4%)	
≥ 2010	5137 (55.2%)	3585 (55.1%)	1552 (55.6%)	
Site:				0.100
Poximal	6802 (73.1%)	4803 (73.8%)	1999 (71.6%)	
Distal	2383 (25.6%)	1629 (25.0%)	754 (27.0%)	
Others	117 (1.26%)	79 (1.21%)	38 (1.36%)	
Grade:				0.689
Grade I	618 (6.64%)	436 (6.70%)	182 (6.52%)	
Grade II	6532 (70.2%)	4549 (69.9%)	1983 (71.0%)	
Grade III	1828 (19.7%)	1299 (20.0%)	529 (19.0%)	
Grade IV	324 (3.48%)	227 (3.49%)	97 (3.48%)	
Histology Type:				0.199
Adenocarcinoma	8343 (89.7%)	5834 (89.6%)	2509 (89.9%)	
Mucinous adenocarcinoma	769 (8.27%)	533 (8.19%)	236 (8.46%)	
Others	190 (2.04%)	144 (2.21%)	46 (1.65%)	
Tumor Size:				0.961
< 5 cm	5394 (58.0%)	3774 (58.0%)	1620 (58.0%)	
≥5 cm	3908 (42.0%)	2737 (42.0%)	1171 (42.0%)	
Marital Status:				0.305
Married	4803 (51.6%)	3372 (51.8%)	1431 (51.3%)	
Single	831 (8.93%)	566 (8.69%)	265 (9.49%)	
Widowed	2892 (31.1%)	2045 (31.4%)	847 (30.3%)	
Separated/Divorced	776 (8.34%)	528 (8.11%)	248 (8.89%)	
CEA:				0.535
Negative/Normal	5825 (62.6%)	4091 (62.8%)	1734 (62.1%)	
Positive/elevated	3477 (37.4%)	2420 (37.2%)	1057 (37.9%)	
T stage:				0.282
T1	743 (7.99%)	513 (7.88%)	230 (8.24%)	
T2	1553 (16.7%)	1086 (16.7%)	467 (16.7%)	
T3	5697 (61.2%)	4022 (61.8%)	1675 (60.0%)	
T4	1309 (14.1%)	890 (13.7%)	419 (15.0%)	
N stage:				0.576
N0	5899 (63.4%)	4141 (63.6%)	1758 (63.0%)	
N1	2290 (24.6%)	1606 (24.7%)	684 (24.5%)	
N2	1113 (12.0%)	764 (11.7%)	349 (12.5%)	

Table 1.

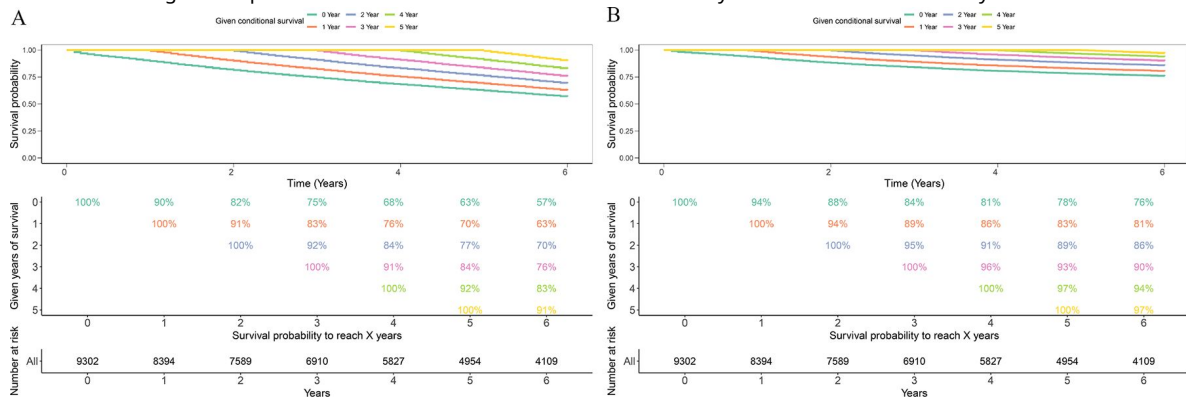
Baseline characteristics of enrolled patients

Table 1. (continued)

Radiation				0.177
None/Unknown	9228 (99.2%)	6465 (99.3%)	2763 (99.0%)	
Yes	74 (0.80%)	46 (0.71%)	28 (1.00%)	
Chemotherapy:				0.468
None/Unknown	7206 (77.5%)	5030 (77.3%)	2176 (78.0%)	
Yes	2096 (22.5%)	1481 (22.7%)	615 (22.0%)	

Figure 1.

Conditional survival analysis of elderly non-metastatic colon cancer patients who undergo colectomy. (A) Kaplan-Meier curves estimating overall survival for individuals who had successfully survived a duration of 0-6 years. (B) Kaplan-Meier curves estimating colon-specific survival for individuals who had successfully survived a duration of 0-6 years.



Variables		At baseline	At 1 year	At 2 years	At 3 years	At 4 years	At 5 years
Total		62.5	63.2	62.8	62.1	61.6	59.8
Race	White	62.4	63.0	62.2	61.4	60.6	58.8
	Black	56.9	58.1	59.5	56.7	58.0	57.2
	Others	65.6	66.4	67.4	68.1	68.8	66.7
	d (Black vs. White)	-0.114	-0.102	-0.056	-0.097	-0.053	-0.032
	d (Others vs. White)	0.066	0.071	0.108	0.138	0.167	0.161
Sex	Male	60.2	60.3	59.1	58.6	58.3	58.1
	Female	64.3	65.3	65.6	64.6	64.0	61.0
	d (Female vs. Male)	0.085	0.104	0.134	0.124	0.117	0.059
Year of diagnosis	< 2010	61.2	62.3	62.3	62.6	61.9	60.0
	≥2010	63.5	63.9	63.3	61.3	61.1	60.3
	d (≥2010 vs. < 2010)	0.048	0.033	0.021	-0.028	-0.016	0.006
Site	Poximal	62.6	63.4	63.5	62.5	62.3	60.1
	Distal	63.2	63.3	61.5	61.3	60.0	59.1
	Others	0.414	47.8	49.0	48.7	55.2	59.0
	d (Distal vs. Poximal)	0.012	-0.002	-0.041	-0.025	-0.047	-0.020
	d (Others vs. Poximal)	-0.438	-0.323	-0.300	-0.284	-0.146	-0.022
Grade	Grade I	67.4	66.5	66.2	63.9	62.7	61.1
	Grade II	64.3	63.8	62.8	62.3	62.0	60.4
	Grade III	56.0	60.0	62.0	61.0	60.4	57.5
	Grade IV	53.3	60.9	59.3	58.5	55.5	57.2
	d (Grade II vs. Grade I)	-0.064	-0.056	-0.070	-0.033	-0.014	-0.014
	d (Grade III vs. Grade I)	-0.235	-0.135	-0.087	-0.060	-0.047	-0.073
	d (Grade IV vs. Grade I)	-0.291	-0.116	-0.143	-0.111	-0.148	-0.080
Histology type	Adenocarcinoma	63.0	63.6	63.1	62.3	62.2	60.7
	Mucinous adenocarcinoma	58.9	58.7	59.4	57.8	54.5	51.5
	Others	54.9	61.3	63.5	70.5	70.6	54.0
	d (Mucinous adenocarcinoma vs. Adenocarcinoma)	-0.085	-0.102	-0.077	-0.093	-0.158	-0.188
	d (Others vs. Adenocarcinoma)	-0.167	-0.048	0.008	0.169	0.173	-0.137
Tumor size	< 5cm	65.5	65	63.5	63.3	63.5	61.8
	≥5cm	58.4	60.5	61.7	60.2	58.7	56.7

Table 2.

Five-year conditional overall survival rates (%).

	d (≥ 5 cm vs. < 5 cm)	-0.147	-0.093	-0.037	-0.064	-0.099	-0.104
Marital status	Married	65.7	66.9	66.1	65.8	65.3	63.6
	Single	59.3	60.3	62.6	63.9	64.4	63.8
	Widowed	58.7	58.3	58	55.9	55.0	53.0
	Separated/Divorced	60.2	61.0	59.9	59.4	59.4	56.8
	d (Single vs. Married)	-0.132	-0.137	-0.072	-0.039	-0.019	0.004
	d (Widowed vs. Married)	-0.145	-0.178	-0.168	-0.204	-0.212	-0.216
CEA	d (Separated/Divorced vs. Married)	-0.114	-0.122	-0.128	-0.132	-0.121	-0.139
	Negative/Normal	68.4	67.8	66.5	64.8	64.2	62.4
	Positive/Elevated	52.6	54.8	55.7	56.6	55.9	53.9
T stage	d (Positive/Elevated vs. Negative/Normal)	-0.326	-0.270	-0.223	-0.169	-0.171	-0.173
	T1	77.8	76.6	73.9	69.4	67.8	65.0
	T2	75.0	73.6	69.6	66.2	64.1	61.5
	T3	62.1	61.8	61.4	61.3	60.9	59.4
	T4	40.6	46.1	50.0	51.8	54.0	53.1
	d (T2 vs. T1)	-0.058	-0.062	-0.089	-0.066	-0.076	-0.071
	d (T3 vs. T1)	-0.324	-0.307	-0.259	-0.167	-0.142	-0.114
	d (T4 vs. T1)	-0.768	-0.632	-0.494	-0.363	-0.284	-0.243
N stage	N0	70.0	69.0	66.4	64.1	62.8	60.1
	N1	54.9	56.4	58.1	58.9	58.9	58.3
	N2	38.3	42.6	48.3	53.9	58.5	61.2
	d (N1 vs. N0)	-0.312	-0.261	-0.172	0.107	-0.080	-0.037
	d (N2 vs. N0)	-0.655	-0.547	-0.374	-0.210	-0.088	0.022
Radiation	None/Unknown	62.6	63.2	62.8	62.1	61.7	59.9
	Yes	47.3	52.8	58.3	56.1	44.5	45.7
	d (Yes vs. None/Unknown)	-0.316	-0.216	-0.093	-0.124	-0.354	-0.290
Chemotherapy	None/Unknown	62.4	63.0	62.0	60.5	59.8	57.5
	Yes	62.9	63.7	65.6	67.4	67.7	67.9
	d (Yes vs. None/Unknown)	0.010	0.015	0.074	0.142	0.162	0.212

Table 2. (continued)

Variables		At baseline	At 1 year	At 2 years	At 3 years	At 4 years	At 5 years
Total		78.1	80.9	84.2	86.9	89.3	90.9
Race	White	78.3	81.5	84.8	87.3	89.5	90.8
	Black	73.1	74.5	79.1	82.5	86.5	89.5
	Others	78.4	80.0	82.2	86.2	89.4	90.4
	d (Black vs. White)	-0.126	-0.178	-0.156	-0.142	-0.097	-0.045
	d (Others vs. White)	0.002	-0.038	-0.071	-0.033	-0.003	-0.014
Sex	Male	77.8	80.0	82.4	84.7	87.7	89.3
	Female	78.3	81.6	85.4	88.5	90.5	91.6
	d (Female vs. Male)	0.012	0.041	0.082	0.113	0.090	0.080
Year of diagnosis							
	< 2010	77.1	80.1	83.8	87.0	89.3	90.9

Table 3.

Five-year conditional colon-specific survival rates (%).

	≥2010	78.9	81.7	84.4	86.7	89.2	89.6
	d (≥2010 vs. < 2010)	0.044	0.041	0.016	-0.009	-0.003	-0.045
Site	Poximal	78.1	81.5	85.6	88.2	90.7	91.9
	Distal	78.3	79.7	80.6	83.7	86.0	87.5
	Others	68.1	74.9	77.6	82.7	85.5	91.2
	d (Distal vs. Poximal)	0.005	-0.046	-0.137	-0.133	-0.152	-0.153
Grade	d (Others vs. Poximal)	-0.242	-0.168	-0.219	-0.163	-0.168	-0.024
	Grade I	85.6	86.2	87.9	89.1	91.3	92.6
	Grade II	80.0	81.5	84.0	86.7	89.4	90.7
	Grade III	70.5	77.4	83.6	86.7	88.1	89.2
Histology type	Grade IV	67.2	77.2	82.4	87.7	89.7	94.3
	d (Grade II vs. Grade I)	-0.135	-0.120	-0.107	-0.071	-0.061	-0.066
	d (Grade III vs. Grade I)	-0.365	-0.224	-0.118	-0.071	-0.104	-0.118
	d (Grade IV vs. Grade I)	-0.445	-0.229	-0.151	-0.041	-0.052	0.059
	Adenocarcinoma	78.5	81.2	84.3	86.9	89.3	90.8
	Mucinous adenocarcinoma	76.3	78.7	83.5	86.9	89.3	88.6
	Others	64.6	76.0	82.1	91.3	92.3	NA
	d (Mucinous adenocarcinoma vs. Adenocarcinoma)	-0.053	-0.064	-0.022	0	0	-0.076
Tumor size	d (Others vs. Adenocarcinoma)	-0.336	-0.132	-0.060	0.130	0.097	NA
	< 5cm	81.3	82.7	84.8	87.5	89.7	90.8
	≥5cm	73.5	78.3	83.1	86.0	88.8	90.4
	d (≥5cm vs. < 5cm)	-0.189	-0.112	-0.047	-0.044	-0.029	-0.014
Marital status	Married	79.5	82.0	84.5	87.0	89.7	91.6
	Single	75.3	77.4	80.9	84.7	88.3	90.2
	Widowed	76.8	79.8	84.1	87.2	89.2	90.0
	Separated/Divorced	76.9	82.2	85.6	87.5	87.7	86.7
CEA	d (Single vs. Married)	-0.101	-0.117	-0.099	-0.068	-0.045	-0.014
	d (Widowed vs. Married)	-0.065	-0.056	-0.011	0.006	-0.016	-0.028
	d (Separated/Divorced vs. Married)	-0.063	0.005	0.030	0.015	-0.065	-0.143
	Negative/Normal	83.8	85.4	87.6	89.5	91.1	92.0
T stage	Positive/Elevated	68.1	72.6	77.5	81.7	85.4	87.4
	d (Positive/Elevated vs. Negative/Normal)	-0.380	-0.326	-0.277	-0.231	-0.184	-0.160
	T1	94.2	95.0	94.7	94.4	94.4	94.4
	T2	92.7	93.4	93.5	93.6	93.9	93.8
N stage	T3	78	79.9	83.1	85.9	88.3	89.8
	T4	50.8	58.2	65.7	72.7	80.5	84.8
	d (T2 vs. T1)	-0.036	-0.041	-0.033	-0.024	-0.016	-0.021
	d (T3 vs. T1)	-0.392	-0.384	-0.318	-0.252	-0.197	-0.160
	d (T4 vs. T1)	-1.049	-0.936	-0.795	-0.643	-0.450	-0.334
	N0	88.1	89.2	90.2	91.0	92.1	92.4
	N1	67.9	70.8	75.6	80.7	84.5	86.9
	N2	46.2	53.2	62.3	70.0	78.3	85.2
	d (N1 vs. N0)	-0.488	-0.468	-0.400	-0.305	-0.246	-0.191
	d (N2 vs. N0)	-1.013	-0.916	-0.765	-0.622	-0.446	-0.250

Table 3. (continued)

Radiation	None/Unknown	78.2	81.1	84.3	87.0	89.4	90.7
	Yes	57.1	61.1	70.6	78.4	80.1	NA
	d (Yes vs. None/Unknown)	-0.510	-0.509	-0.376	-0.255	-0.301	NA
Chemotherapy	None/Unknown	80.5	83.8	86.7	89.0	91.0	91.6
	Yes	70.2	71.8	75.9	80.2	83.9	87.5
	d (Yes vs. None/Unknown)	-0.249	-0.305	-0.296	-0.261	-0.230	-0.143

Table 3. (continued)

Variables linked to the OS and CSS in patients who have attained a defined duration of survival

Age, race, sex, year of diagnosis, site, grade, marital status, CEA level, T stage, N stage, and chemotherapy were independent prognostic factors for OS at baseline (**Table 4** and **Supplementary Table 1**). The prognostic factors for patients who survived for at least one year differed from those at baseline, including age, race, sex, marital status, CEA levels, T stage, N stage, and chemotherapy. After three years, age, race, sex, histology type, marital status, CEA levels, T stage, N stage, and chemotherapy independently affected OS. At the 5-year mark, age, race, sex, marital status, CEA levels, T stage, and chemotherapy were key predictors of OS, consistent with the baseline and other survival intervals.

Age, year of diagnosis, site, grade, marital status, CEA, T stage, N stage, and chemotherapy were determined to be independent prognostic factors for cancer-specific survival (CSS) at baseline, as indicated in **Table 5** and **Supplementary Table 2**. After a minimum of one year of survival, age, CEA levels, T stage, N stage, and chemotherapy remained significant risk factors for CSS. For patients who survived beyond three years, age, sex, tumor location, marital status, CEA levels, T stage, and N stage emerged as influential factors. Among those who surpassed a five-year milestone, age, tumor site, CEA level, and T stage were independent prognostic factors. Constant predictors of CSS for newly diagnosed patients and 1-, 3-, and 5-year survivors were age, CEA, and T stage.

Conditional nomogram construction and validation

Figure 2 and **3** illustrate the OS and CSS nomograms developed for cancer patients at different stages of their journey, including the start and at the one-year, three-year, and five-year milestones. These nomograms consider the specific risk factors for each individual. Our investigation delved deeper into the performances of these nomograms. Baseline analysis of the OS nomogram in the training set revealed an area under the curve (AUC) of 0.703, 0.713, and 0.713 for the one-year, three-year, and five-year intervals, respectively (**Figure 4**). In the test set, the corresponding AUC were 0.714, 0.722, and 0.719, respectively (**Supplementary Figure 2**). For patients who survived for at least one year, the OS nomogram showed AUC of 0.700, 0.697, and 0.698 in the training set and 0.698, 0.694, and 0.701 in the test set, respectively, for the same intervals. For patients surviving for 3 years, the OS nomogram achieved AUC of 0.652, 0.665, and 0.694 in the training set and 0.663, 0.666, and 0.688 in the test set, respectively. Additionally, for patients surviving 5 years, the training set's OS nomogram yielded AUC of 0.661, 0.694, and 0.695 for the one-year, three-year, and five-year durations. In the test set, AUC were 0.590, 0.674, and 0.681 for the same intervals. Regarding the CSS nomogram, at baseline, the training cohort demonstrated AUC values of 0.771, 0.788, and 0.782 for the one-year, three-year, as well as five-year periods (**Figure 5**), while the test cohort achieved corresponding AUC values of 0.787, 0.804, and 0.800, respectively (**Supplementary Figure 3**). For patients who survived for 1 year, the training cohort CSS nomogram attained AUC of 0.785, 0.795, and 0.781 for the one-year, three-year, and five-year intervals, whereas the test cohort displayed AUC of 0.780, 0.778, and 0.761, respectively. Among patients surviving 3 years, the training cohort showed AUC of 0.765, 0.732, and 0.727 over the one-year, three-year, and five-year periods, whereas the test cohort displayed AUC of 0.786, 0.732, and 0.697, respectively, for the same duration. For patients who survived for 5 years, the training cohort CSS nomogram achieved AUC values of 0.664, 0.669, and 0.682 for the one-year, three-year, and five-year periods, whereas the test cohort showed AUC values of 0.585, 0.629, and 0.655, respectively.

The calibration curves of the OS and CCS nomograms when utilized for these training and validation sets are also presented. The outcomes demonstrated strong concordance between anticipated and observed survival rates in both the training and validation sets, indicating the

Table 4.

Multivariate analyses of overall deaths in baseline, $\geq$ one-, three-, five-year survivors in the training cohort.

Variables		Baseline survivors		> 1 year survivors		> 3 year survivors		> 5 year survivors	
		HR	p	HR	p	HR	p	HR	p
Age		1.07 (1.06-1.08)	<0.001	1.08 (1.07-1.08)	<0.001	1.09 (1.08-1.10)	<0.001	1.10 (1.09-1.12)	<0.001
Race	White	Ref		Ref		Ref			
	Black	1.20 (1.04-1.38)	0.010	1.10 (0.95-1.28)	0.217	1.06 (0.87-1.29)	0.579	1.14 (0.90-1.44)	0.287
	Others	1.20 (1.04-1.38)	<0.001	0.77 (0.69-0.85)	<0.001	0.77 (0.67-0.87)	<0.001	0.69 (0.62-0.77)	<0.001
Sex	Male	Ref		Ref		Ref			
	Female	0.73 (0.68-0.78)	<0.001	0.71 (0.66-0.77)	<0.001	0.67 (0.61-0.74)	<0.001	0.69 (0.62-0.77)	<0.001
Year of diagnosis									
		< 2010							
		≥ 2010	0.90 (0.84-0.98)	0.004					
Site	Poximal	Ref							
	Distal	1.04 (0.97-1.12)	0.346						
	Others	1.52 (1.16-2.00)	0.003						
Grade	Grade I	Ref		Ref					
	Grade II	1.04 (0.91-1.19)	0.537	0.96 (0.83-1.10)	0.524				
	Grade III	1.17 (1.01-1.35)	0.035	0.97 (0.83-1.13)	0.672				
	Grade IV	1.21 (0.98-1.50)	0.074	0.97 (0.76-1.24)	0.672				
Histology type	Adenocarcinoma	Ref		Ref		Ref		Ref	
	Mucinous adenocarcinoma	1.04 (0.93-1.16)	0.462	1.04 (0.92-1.18)	0.494	1.06 (0.92-1.23)	0.393	1.09 (0.92-1.30)	0.311
	Others	0.81 (0.64-1.01)	0.064	0.90 (0.68-1.18)	0.452	0.65 (0.45-0.95)	0.025	0.80 (0.52-1.22)	0.297
Tumor size									
		< 5cm							
		≥ 5 cm	Ref	Ref				Ref	
			1.01 (0.94-1.08)	0.794	0.96 (0.89-1.04)	0.324		1.03 (0.92-1.14)	0.608
Marital status	Married	Ref		Ref		Ref		Ref	
	Single	1.09 (0.97-1.23)	0.152	1.12 (0.98-1.28)	0.089	1.21 (1.03-1.41)	0.02	1.02 (0.83-1.24)	0.874
	Widowed	1.14 (1.05-1.23)	0.001	1.15 (1.05-1.25)	0.002	1.24 (1.12-1.37)	<0.001	1.22 (1.08-1.38)	0.002
	Separated/Divorced	1.45 (1.29-1.64)	<0.001	1.36 (1.19-1.56)	<0.001	1.47 (1.25-1.73)	<0.001	1.57 (1.29-1.92)	<0.001
CEA	Negative/Normal	Ref				Ref			
T stage	Positive/Elevated	1.38 (1.29-1.47)	<0.001	1.35 (1.25-1.45)	<0.001	1.39 (1.26-1.50)	<0.001	1.26 (1.13-1.40)	<0.001
	T1	Ref		Ref		Ref		Ref	
N stage	T2	1.03 (0.88-1.20)	0.746	0.99 (0.84-1.16)	0.880	1.08 (0.90-1.29)	0.419	1.06 (0.86-1.31)	0.586
	T3	1.21 (1.05-1.39)	0.009	1.19 (1.03-1.38)	0.022	1.12 (0.95-1.32)	0.161	1.07 (0.88-1.30)	0.519
	T4	2.02 (1.72-2.38)	<0.001	1.84 (1.54-2.18)	<0.001	1.42 (1.16-1.73)	<0.001	1.31 (1.02-1.70)	0.037
	N0	Ref		Ref		Ref			
	N1	1.47 (1.35-1.59)	<0.001	1.51 (1.38-1.66)	<0.001	1.26 (1.13-1.41)	<0.001		
Chemotherapy	N2	2.30 (2.06-2.56)	<0.001	2.27 (2.00-2.57)	<0.001	1.64 (1.40-1.93)	<0.001		
	None/Unknown	Ref		Ref		Ref		Ref	
	Yes	0.65 (0.59-0.71)	<0.001	0.65 (0.59-0.73)	<0.001	0.79 (0.70-0.90)	<0.001	0.83 (0.72-0.96)	0.009
HR Hazard ratio									

Table 5.

Multivariate analyses of colon-specific deaths in baseline, $\geq$ one-, three-, five-year survivors in the training cohort.

Variables		Baseline survivors		> 1 year survivors		> 3 year survivors		> 5 year survivors	
		HR	p	HR	p	HR	p	HR	p
Age		1.03 (1.02-1.04)	<0.001	1.04 (1.03-1.05)	<0.001	1.04 (1.02-1.05)	<0.001	1.06 (1.03-1.08)	<0.001
Race	White			Ref		Ref			
	Black			1.26 (1.00-1.60)	0.052	1.32 (0.95-1.84)	0.095		
Sex	Others			0.88 (0.75-1.05)	0.148	0.93 (0.73-1.18)	0.552		
	Male					Ref			
	Female					0.70 (0.59-0.84)	<0.001		
Year of diagnosis	< 2010	Ref							
	$\geq$ 2010	0.84 (0.76-0.93)	0.001						
Site	Poximal	Ref				Ref		Ref	
	Distal	1.09 (0.97-1.23)	0.149			1.43 (1.20-1.71)	<0.001	1.71 (1.34-2.20)	<0.001
	Others	1.79 (1.26-2.56)	0.001			0.83 (0.37-1.86)	0.649	1.32 (0.42-4.17)	0.632
Grade	Grade I	Ref		Ref					
	Grade II	1.15 (0.90-1.47)	0.266	1.00 (0.77-1.30)	0.981				
	Grade III	1.36 (1.05-1.77)	0.020	0.97 (0.73-1.28)	0.816				
	Grade IV	1.32 (0.94-1.85)	0.109	0.81 (0.54-1.21)	0.300				
Histology type	Adenocarcinoma								
	Mucinous adenocarcinoma	0.96 (0.80-1.15)	0.648						
	Others	0.94 (0.70-1.28)	0.710						
Tumor size	< 5cm	Ref		Ref					
	$\geq$ 5cm	0.96 (0.87-1.07)	0.499	0.90 (0.80-1.02)	0.096				
Marital status	Married	Ref		Ref		Ref			
	Single	1.10 (0.92-1.32)	0.282	1.25 (1.02-1.52)	0.029	1.41 (1.07-1.86)	0.014		
	Widowed	0.99 (0.88-1.11)	0.842	0.98 (0.86-1.13)	0.799	1.15 (0.93-1.41)	0.195		
	Separated/Divorced	1.32 (1.10-1.59)	0.002	1.02 (0.81-1.28)	0.888	1.29 (0.95-1.76)	0.108		
CEA	Negative/Normal	Ref		Ref		Ref		Ref	
	Positive/Elevated	1.55 (1.40-1.72)	<0.001	1.48 (1.32-1.67)	<0.001	1.71 (1.45-2.01)	<0.001	1.41 (1.11-1.81)	0.006
T stage	T1	Ref		Ref		Ref		Ref	
	T2	0.98 (0.69-1.40)	0.903	1.04 (0.70-1.55)	0.844	1.31 (0.78-2.20)	0.299	1.41 (0.75-2.65)	0.281
	T3	1.98 (1.45-2.70)	<0.001	2.18 (1.54-3.09)	<0.001	2.35 (1.49-3.71)	<0.001	1.88 (1.06-3.32)	0.030
	T4	4.27 (3.09-5.91)	<0.001	4.34 (3.01-6.26)	<0.001	3.97 (2.44-6.47)	<0.001	2.77 (1.45-5.28)	0.002
N stage	N0	Ref		Ref		Ref			
	N1	2.36 (2.08-2.68)	<0.001	2.60 (2.23-3.02)	<0.001	1.82 (1.48-2.25)	<0.001	1.36 (0.99-1.87)	0.056
	N2	4.45 (3.84-5.16)	<0.001	4.89 (4.11-5.83)	<0.001	2.94 (2.28-3.79)	<0.001	1.39 (0.89-2.18)	0.148
Radiation	None/Unknown			Ref					
	Yes			1.52 (0.95-2.43)	0.078				
Chemotherapy	None/Unknown	Ref		Ref		Ref		Ref	
	Yes	0.70 (0.62-0.80)	<0.001	0.74 (0.63-0.86)	<0.001	0.94 (0.76-1.17)	0.596	1.12 (0.80-1.58)	0.498
HR Hazard ratio									

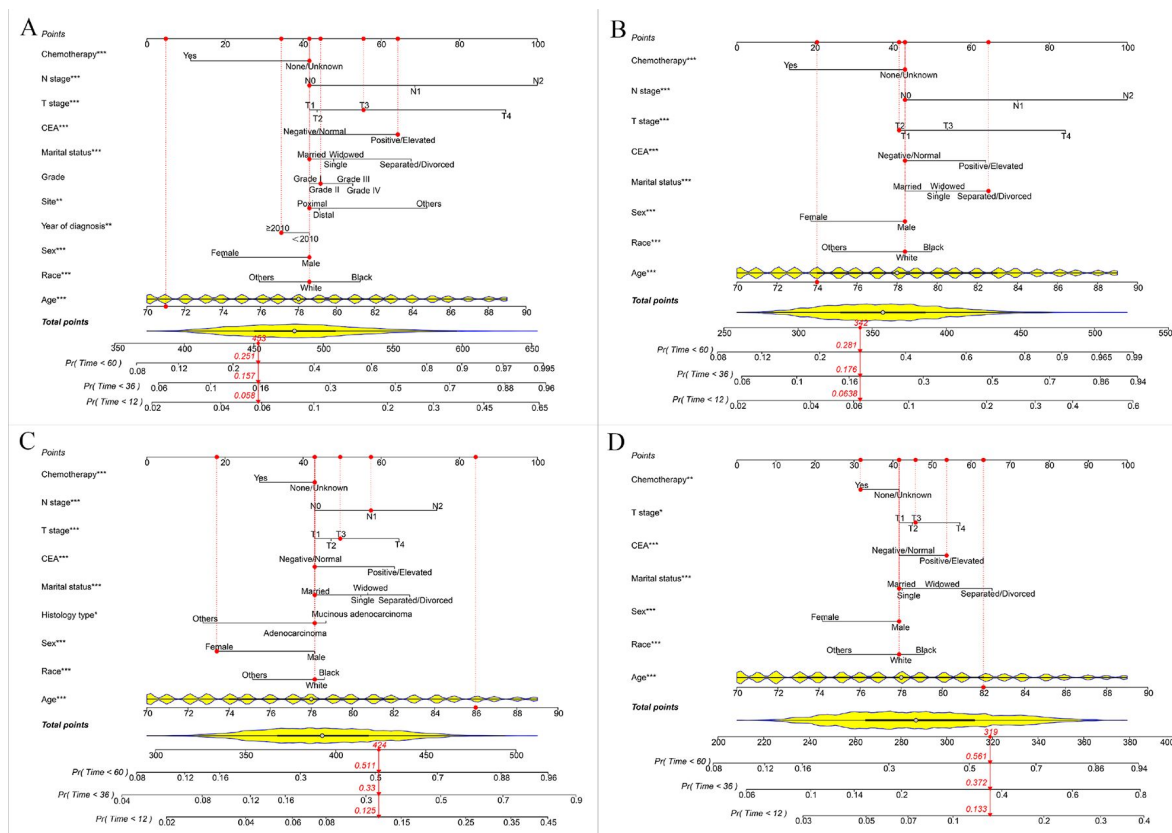


Figure 2.

Nomograms to predict one-, three-, and five-overall survival for baseline (A), one-year (B), three-year (C), and five-year (D) survivors.

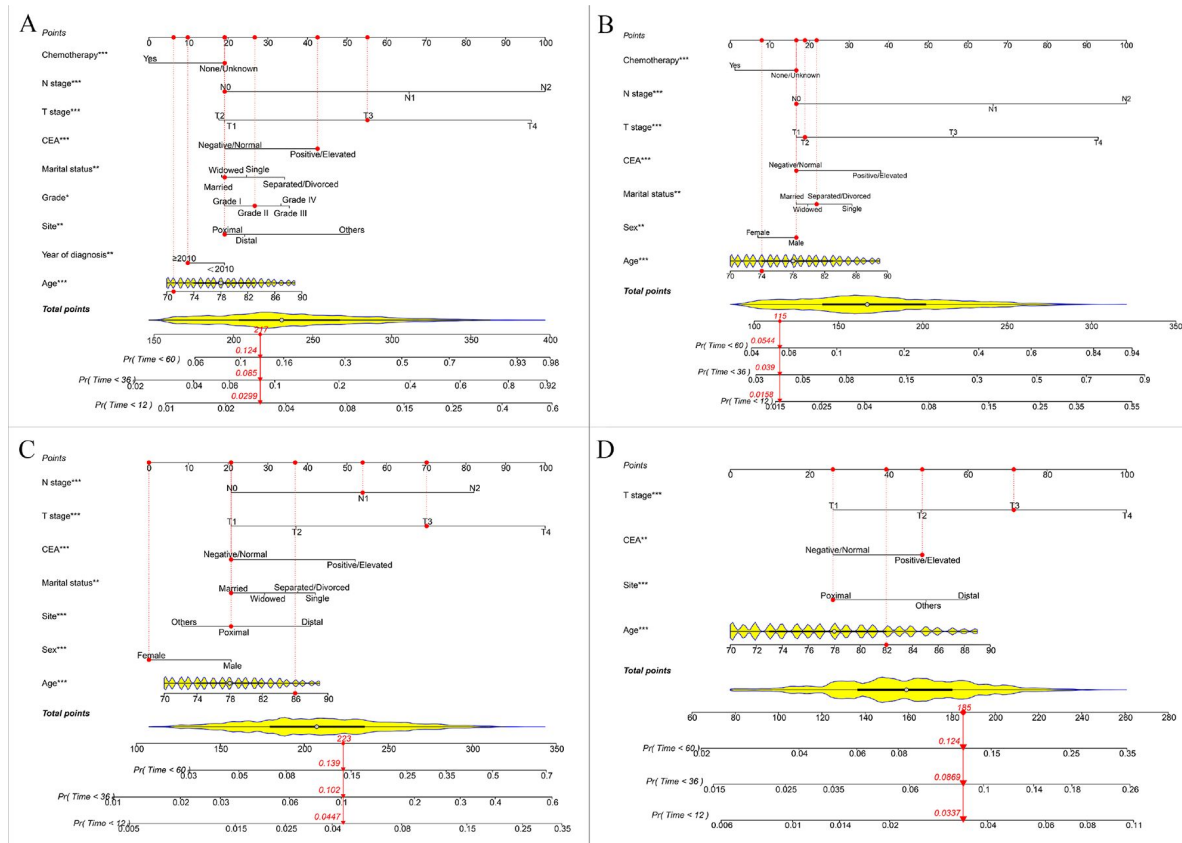


Figure 3.

Nomograms to predict one-, three-, and five-year colon-specific survival for baseline (A), one-year (B), three-year (C), and five-year (D) survivors.

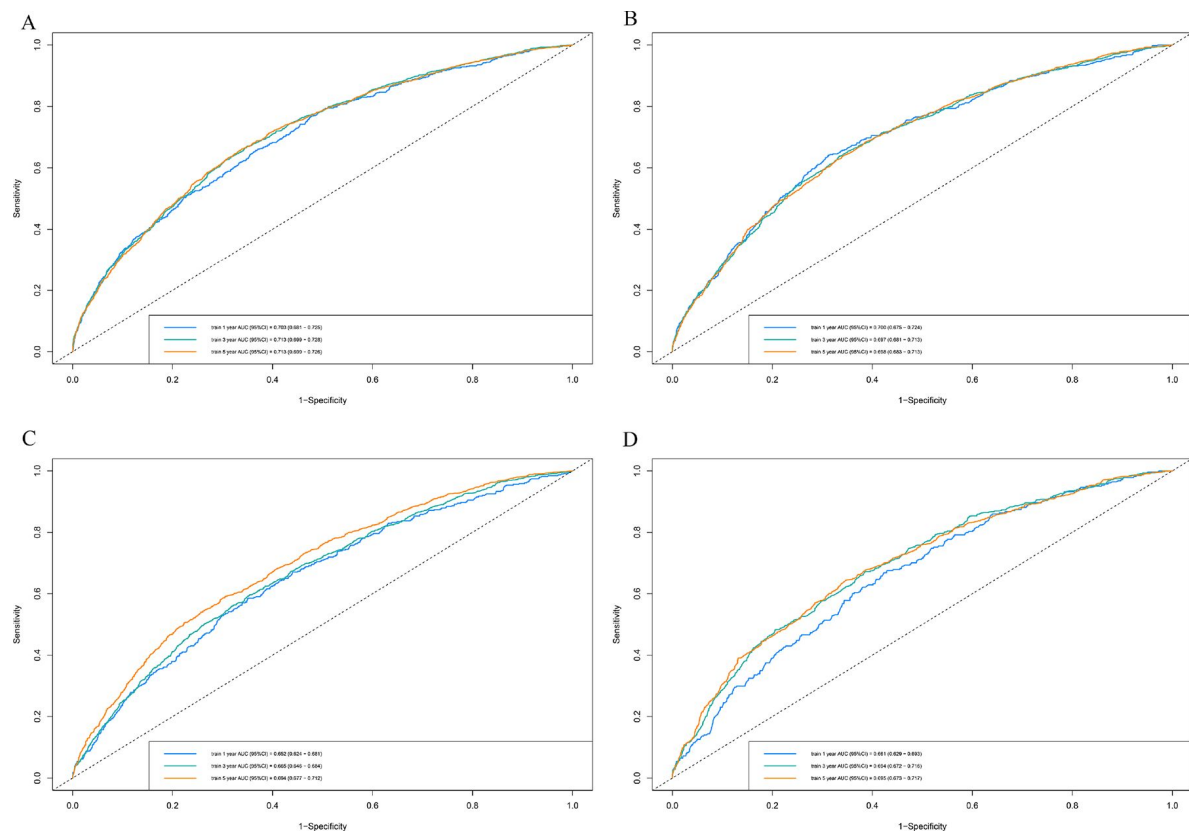


Figure 4.

ROC curve analyses in the training cohort for one-, three-, and five-year overall survival in baseline (A), one-year (B), three-year (C), and five-year (D) survivors. ROC, receiver operator characteristic curve; AUC, area under the curve.

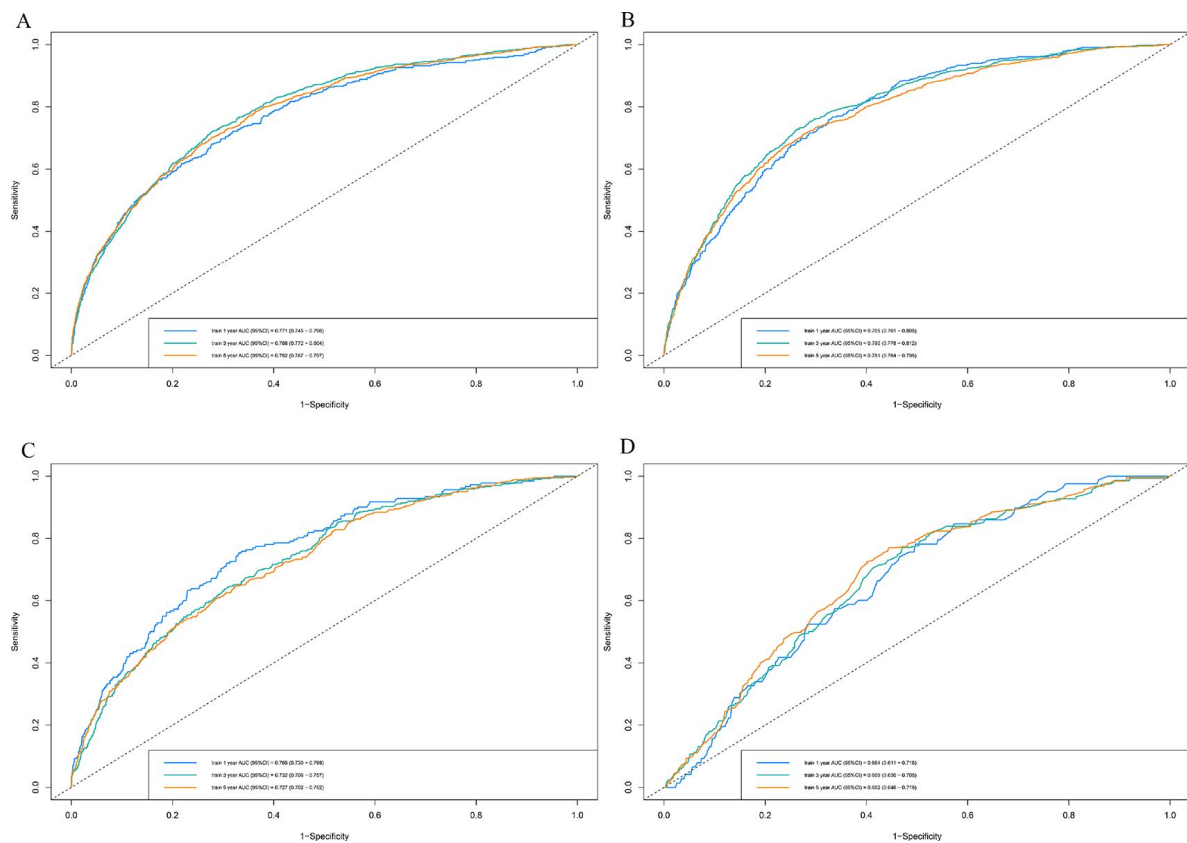


Figure 5.

ROC curve analyses in the training cohort for one-, three-, and five-year colon-specific survival in baseline (A), one-year (B), three-year (C), and five-year (D) survivors. ROC, receiver operator characteristic curve; AUC, area under the curve.

accuracy of these nomograms in predicting survival probabilities (**Supplementary Figure 4**⁴[↗](#)⁵[↗](#)). In addition, the DCA curves also indicated the favorable clinical utility of these nomograms (**Supplementary Figure 8**⁸[↗](#)¹¹[↗](#)).

4. Discussion

In this investigation, we examined the survival data of elderly patients with non-metastatic colon cancer who had undergone colectomy in detail. The uniqueness and notable attributes of our research encompass the following: we conducted an analysis of the CS rates, allowing for a more precise assessment of OS and CSS outcomes in both medium-term and long-term survivors; this analysis aids in accurately forecasting survival rates and advancing personalized medical approaches; our findings revealed the nomograms to possess favorable discriminatory and predictive capabilities when forecasting OS and CSS rates.

In the coming years, we expect a higher incidence of colon cancer in the elderly population. This can be attributed to a longer life expectancy and improved rates of survival¹⁸[↗](#). Colectomy is the primary curative treatment for the elderly. Previous studies have reported a notable correlation between surgical resection and improved prognosis in older patients with colon cancer⁸[↗](#)⁹[↗](#)¹⁹[↗](#). However, the prognosis of elderly patients differs from that of younger individuals owing to the presence of other health conditions, functional disabilities, more advanced tumors, and a decreased ability to maintain internal balance. Elderly patients experience higher rates of complications and mortality following surgery as well as a less favorable overall outlook³[↗](#)²⁰[↗](#). Given the complexities surrounding older adults with colon cancer, there is an urgent need for further research focusing on this demographic, particularly on those who have undergone surgical procedures. Currently, there is limited information on the medium- and long-term postoperative prognoses of elderly patients with colon cancer, and few studies have investigated the practical application of prediction models specifically tailored to this population. Wang²¹[↗](#) et al. developed a prognostic model that accurately predicted the outcomes of elderly patients with colorectal cancer. However, their study included patients with and without metastatic disease, and it is widely recognized that prognosis differs significantly between these two groups. Additionally, their study focused only on CSS and not OS. To our knowledge, this study is the first to investigate the development and validation of a reliable nomogram for predicting both OS and CSS in elderly adults with non-metastatic colon cancer who have undergone colectomy. This model is expected to provide patients and healthcare providers with a valuable tool for guiding follow-up care and treatment decisions.

CS provides a precise estimation of the likelihood of a specific time point in the course of a disease, closely tied to individual long-term survival rates based on patients' time of survival¹¹[↗](#). This allows for a more accurate depiction of prognostic outcomes. To our knowledge, this study is the first to show CS in elderly individuals with non-metastatic colon cancer who have undergone colectomy. In this study, survival analysis revealed an upward trend in actuarial OS and CSS among elderly patients as their survival duration increased annually. However, COS and CCSS rates showed contrasting patterns. Initially, the COS5 rates experienced a slight increase, followed by a subsequent decrease, whereas the CCSS5 rates demonstrated a consistent upward trend over the five-year duration. Compared with the previously noted trend of annual increases in the likelihood of COS rates for malignant tumors¹¹[↗](#)¹³[↗](#), particularly CRC, this study revealed a distinctive pattern of change in COS rates. Our hypothesis attributed this variance to disparity in the cohort composition. This study differs from previous research by exclusively focusing on older adults instead of encompassing individuals across all age groups. Throughout the course of follow-up, the impact of non-colon cancer-related death determinants accumulated at various stages of the life course, leading to a growing risk of other conditions such as cardiovascular and cerebrovascular diseases among elderly patients with colon cancer. a recent study²²[↗](#), it was found that the majority of deaths in patients aged > 80 years during the follow-up period were

unrelated to CRC. Non-CRC-related causes of death, such as respiratory failure, particularly aspiration pneumonia, cardiovascular disease, and sepsis, were the primary contributors to mortality. Another study suggested that cerebrovascular-specific hazard ratios were 3.10-, 6.67, and 10.95 times greater for CRC patients aged 50-64, 65-74, and over 75 years, respectively, compared to those under 50 years old²³. One of the foremost concerns for those who survive cancer is the risk of mortality. Although the risk of death due to colon cancer surpasses that of other illnesses in the initial years, this specific risk diminishes over time in a discernible trend. Our research showed that over the span of five years, the CCSS5 showed a consistent increase, with the baseline and subsequent conditional survival rates at years 1 through 5 documented as 78.1%, 80.9%, 84.2%, 86.9%, 89.3%, and 90.9%, respectively, indicating a positive trend. The findings from the CS analysis could instill greater optimism in the patients. For instance, consider a patient who initially has a 5-year CSS of 78.1% at the time of diagnosis. Subsequently, through dedicated adherence to treatment and follow-up, the patient successfully achieved 5-year survival, and the CS analysis conveyed the encouraging news: “Your 5-year CSS rate has risen to 90.9%.” On the other hand, in contrast to the general population’s COS outcomes, the comparatively less positive COS outcomes in the elderly indicate a greater need for physicians to prioritize non-colonic causes of death as patients age. Subgroup analysis was performed to further understand the influence of different variables on CS. The rates of COS5 and CCSS5 expression varied in these subgroups. Over time, the variation in the rates of both COS5 and CCSS5 among many subgroups gradually decreased (black vs. White, Grade IV vs. Grade I, Single vs. Married, Positive/Elevated vs. Negative/Normal, T3 vs. T1, T4 vs. T1, N1 vs. N0, and N2 vs. N0). One possible explanation is that patients in each subgroup with some high-risk factors died over time, whereas the remaining patients had a better prognosis, resulting in a smaller difference in survival between the subgroups. The change patterns of COS5 and CCSS5 differed among the subgroups, which may be influenced by multiple factors, such as those affecting cardiovascular death. Future studies of CS in these subgroups should be conducted.

Our study examined the impact of various baseline prognostic factors on patients with different survival durations. Multiple factors, including age, race, sex, year of diagnosis, site, grade, marital status, carcinoembryonic antigen (CEA) levels, T stage, N stage, and chemotherapy, significantly influenced OS according to multifactor analysis. Upon reaching the critical 5-year survival milestone, we identified several key independent prognostic factors that consistently influenced the patient outcomes. These factors included age, race, sex, marital status, CEA levels, T stage, and chemotherapy, demonstrating a predictive value for OS across baseline and 1-year, 3-year, and 5-year intervals. Furthermore, we observed that at the beginning of the study, age, year of diagnosis, site, grade, marital status, CEA levels, T stage, N stage, and chemotherapy independently influenced CSS. However, as time progressed, only age, CEA level, and T stage consistently emerged as predictive factors for CSS in both newly diagnosed patients and those who survived for 1, 3, and 5 years. These variables, as observed in previous studies, could predict OS and CSS at the beginning of diagnosis^{15,19,21}, but their predictive significance diminished over time. To provide more accurate prognostic predictions, we utilized these variables to create nomograms for each type of survival, which demonstrated good prediction performance and a correspondingly good match between the predicted and observed survival rates in the calibration plots. The practical application of these nomograms is also supported by their favorable clinical utility, as shown in the DCA curves. However, it should be noted that, as time progressed, the AUC values of the nomograms gradually declined, indicating a decrease in predictive accuracy. This trend can be explained by the diminishing impact of clinicopathological factors on the prognosis over time. Further research is needed to assess the latest clinical and laboratory factors at specific time points to more accurately identify prognostic factors affecting elderly colon cancer survivors. This study had some limitations. First, given the retrospective nature of the study, it inevitably introduced a degree of selection bias owing to its inherent design. Second, obtaining postoperative complications and adjuvant treatment details from the SEER database for all patients is not feasible, potentially impacting the survival prognosis of elderly individuals with colon cancer.

Third, the analysis of CS necessitates extensive data, thus posing challenges for the external validation of these nomograms. Furthermore, the ongoing update of nomograms is essential to accommodate the evolving treatment strategies.

5. Conclusion

We reported the rates of CS in elderly non-metastatic colon cancer patients following colectomy, enhancing our understanding of survival rates in the medium- and long-term. We also created personalized CS nomograms to predict mortality risk at different intervals. These tools aid surgeons and patients in making informed decisions, and contribute to personalized medicine.

Data Availability

All data produced in the present study are available upon reasonable request to the authors.

Ethical approval

The information used in our study was obtained from the SEER database, which is accessible to the public, thus eliminating the need for ethical clearance.

Sources of funding

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Conflict of Interest declaration

The authors declare that they have no affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

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Supplementary Table 1.

Univariate analyses of overall deaths in baseline, ≥ one-, three- and five-year survivors in the training cohort.

Variables		Baseline survivors		> 1 year survivors		> 3 year survivors		> 5 year survivors	
		HR	p	HR	p	HR	p	HR	p
Age		1.08 (1.07-1.08)	<0.001	1.08 (1.07-1.09)	<0.001	1.09 (1.08-1.10)	<0.001	1.11 (1.10-1.12)	<0.001
Race	White	Ref		Ref		Ref		Ref	
	Black	1.16 (1.01-1.34)	0.032	1.09 (0.94-1.27)	0.238	1.03 (0.85-1.25)	0.758	1.08 (0.85-1.36)	0.535
	Others	1.17 (0.76-0.95)	0.002	0.82 (0.74-0.91)	<0.001	0.80 (0.70-0.91)	<0.001	0.73 (0.62-0.85)	<0.001
Sex	Male	Ref							
	Female	0.91 (0.85-0.97)	0.004	0.88 (0.82-0.94)	<0.001	0.86 (0.79-0.93)	<0.001	0.89 (0.80-0.98)	0.024
Year of diagnosis	< 2010	Ref		Ref		Ref		Ref	
	≥2010	0.93 (0.87-1.00)	0.039	0.95 (0.88-1.03)	0.192	0.99 (0.90-1.08)	0.789	0.98 (0.87-1.11)	0.759
Site	Poximal	Ref		Ref		Ref		Ref	
	Distal	0.98 (0.91-1.06)	0.642	0.98 (0.91-1.06)	0.629	1.07 (0.97-1.17)	0.164	1.02 (0.91-1.14)	0.716
	Others	1.57 (1.20-2.06)	0.001	0.74 (0.98-1.87)	0.063	1.52 (1.09-2.12)	0.015	0.95 (0.53-1.72)	0.870
Grade	Grade I	Ref		Ref		Ref		Ref	
	Grade II	1.14 (1.00-1.29)	0.555	1.04 (0.91-1.20)	0.546	0.98 (0.84-1.15)	0.813	0.98 (0.82-1.19)	0.869
	Grade III	1.47 (1.27-1.69)	<0.001	1.20 (1.04-1.40)	0.016	1.05 (0.88-1.26)	0.568	0.97 (0.78-1.20)	0.788
	Grade IV	1.48 (1.21-1.81)	<0.001	1.24 (0.98-1.57)	0.073	0.92 (0.69-1.24)	0.589	0.97(0.68-1.39)	0.882
Histology type	Adenocarcinoma	Ref		Ref		Ref		Ref	
	Mucinous adenocarcinoma	1.20 (1.08-1.34)	<0.001	1.20 (1.06-1.35)	0.003	1.15 (1.00-1.33)	0.049	1.23 (1.04-1.46)	0.018
	Others	1.21 (0.98-1.51)	0.080	1.23 (0.95-1.60)	0.121	1.06 (0.65-1.36)	0.749	1.12 (0.73-1.70)	0.604
Tumor size	< 5cm	Ref		Ref		Ref		Ref	
	≥5cm	1.24 (1.16-1.32)	<0.001	1.18 (1.10-1.26)	<0.001	1.08 (0.99-1.17)	0.074	1.12 (1.01-1.24)	0.031
Marital status	Married	Ref		Ref		Ref		Ref	
	Single	1.09 (0.96-1.22)	0.178	1.10 (0.96-1.25)	0.169	1.15 (0.98-1.34)	0.087	1.00 (0.82-1.22)	0.996
	Widowed	1.32 (1.23-1.41)	<0.001	1.30 (1.21-1.41)	<0.001	1.41 (1.29-1.55)	<0.001	1.48 (1.33-1.66)	<0.001
	Separated/Divorced	1.30 (1.15-1.46)	<0.001	1.23 (1.07-1.40)	0.003	1.26 (1.07-1.47)	0.005	1.34 (1.11-1.64)	0.003
CEA	Negative/Normal	Ref		Ref		Ref		Ref	
	Positive/Elevated	1.53 (1.44-1.63)	<0.001	1.46 (1.36-1.57)	<0.001	1.39 (1.28-1.52)	<0.001	1.27 (1.14-1.41)	<0.001
T stage	T1	Ref		Ref		Ref		Ref	
	T2	1.14 (0.97-1.33)	0.103	1.09 (0.93-1.28)	0.307	1.16 (0.97-1.39)	0.114	1.13 (0.92-1.40)	0.245
	T3	1.56 (1.36-1.78)	<0.001	1.45 (1.26-1.67)	<0.001	1.31 (1.12-1.53)	<0.001	1.20 (0.99-1.44)	0.065
	T4	2.75 (2.36-3.20)	<0.001	2.30 (1.96-2.71)	<0.001	1.66 (1.36-2.01)	<0.001	1.40 (1.10-1.80)	0.007
N stage	N0	Ref		Ref		Ref		Ref	
	N1	1.36 (1.26-1.46)	<0.001	1.34 (1.23-1.45)	<0.001	1.18 (1.07-1.30)	<0.001	1.01 (0.89-1.14)	0.899
	N2	1.97 (1.80-2.16)	<0.001	1.82 (1.64-2.02)	<0.001	1.34 (1.16-1.54)	<0.001	0.82 (0.67-1.01)	0.066
Radiation	None/Unknown	Ref		Ref		Ref		Ref	
	Yes	1.28 (0.92-1.80)	0.146	1.47 (1.05-2.06)	0.026	1.15 (0.71-1.85)	0.565	1.32 (0.75-2.33)	0.335
Chemotherapy	None/Unknown	Ref		Ref		Ref		Ref	
	Yes	0.85 (0.80-0.92)	<0.001	0.86 (0.79-0.94)	<0.001	0.82 (0.74-0.91)	<0.001	0.65 (0.57-0.74)	<0.001

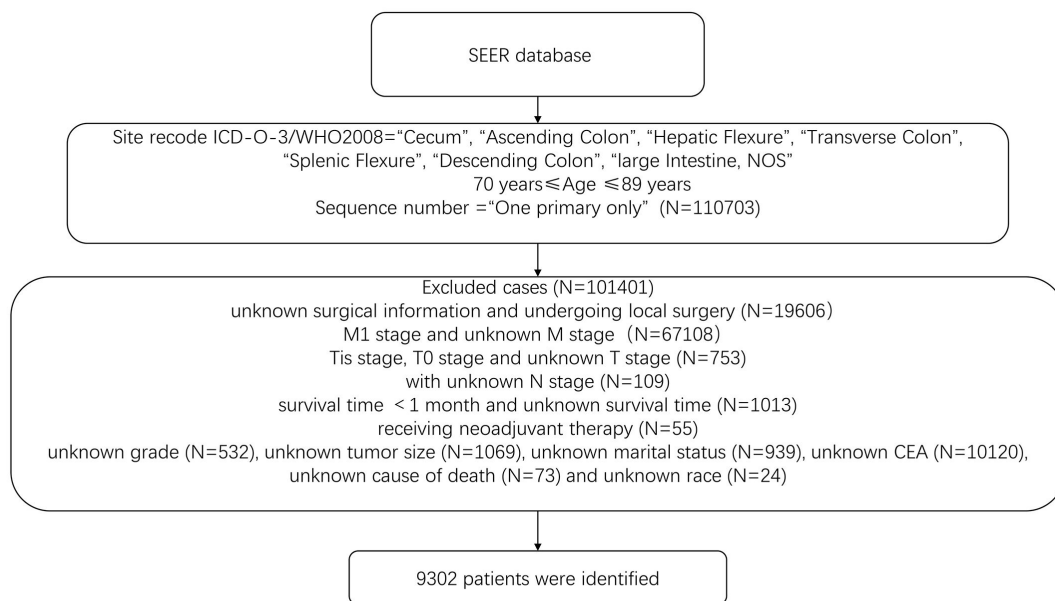
HR Hazard ratio

Supplementary Table 2.

Univariate analysis of colon-specific deaths at baseline, ≥ one-, three-and five-year survivors in the training cohort.

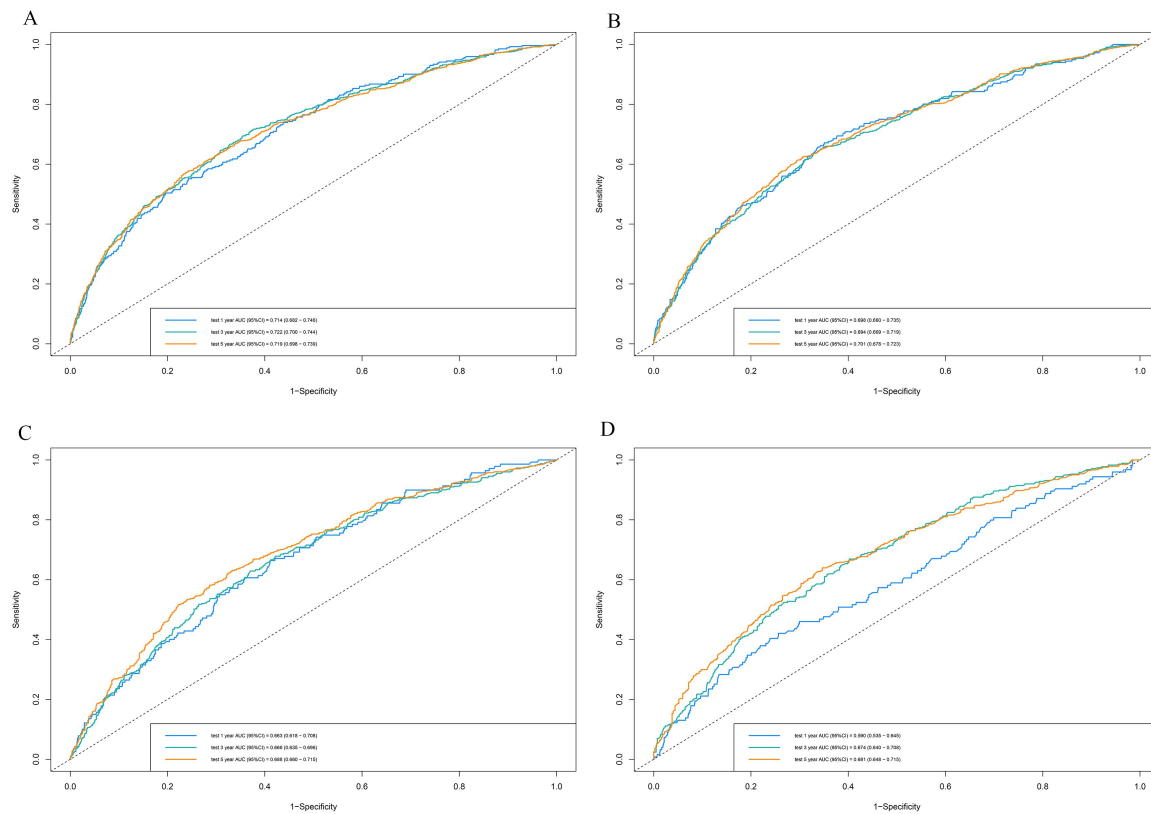
Variables		Baseline survivors		> 1 year survivors		> 3 year survivors		> 5 year survivors	
		HR	p	HR	p	HR	p	HR	p
Age		1.03 (1.03-1.04)	<0.001	1.04 (1.02-1.05)	<0.001	1.03 (1.01-1.04)	<0.001	1.05 (1.03-1.07)	<0.001
Race	White	Ref		Ref		Ref		Ref	
	Black	1.21 (0.98-1.50)	0.077	1.35 (1.07-1.70)	0.011	1.49 (1.08-2.06)	0.015	1.12 (0.65-1.92)	0.688
	Others	1.02 (0.88-1.18)	0.822	1.04 (0.88-1.23)	0.634	1.09 (0.86-1.36)	0.521	0.89 (0.62-1.26)	0.500
Sex	Male	Ref		Ref		Ref		Ref	
	Female	1.01 (0.91-1.11)	0.903	0.95 (0.85-1.07)	0.415	0.81 (0.69-0.95)	0.011	0.87 (0.68-1.11)	0.253
Year of diagnosis	< 2010	Ref		Ref		Ref		Ref	
	≥2010	0.89 (0.81-0.99)	0.033	0.89 (0.79-1.00)	0.051	0.95 (0.80-1.13)	0.586	1.00 (0.76-1.31)	0.993
Site	Proximal	Ref		Ref		Ref		Ref	
	Distal	1.06 (0.94-1.19)	0.341	1.03 (0.91-1.18)	0.621	1.41 (1.19-1.67)	<0.001	1.61 (1.26-2.05)	<0.001
	Others	2.21 (1.55-3.14)	<0.001	1.45 (0.87-2.41)	0.157	1.05 (0.47-2.36)	0.897	1.58 (0.50-4.95)	0.432
Grade	Grade I	Ref		Ref		Ref		Ref	
	Grade II	1.47 (1.15-1.87)	0.002	1.26 (0.98-1.63)	0.074	1.12 (0.81-1.56)	0.488	1.14 (0.71-1.83)	0.584
	Grade III	2.41 (1.87-3.11)	<0.001	1.71 (1.30-2.26)	<0.001	1.13 (0.78-1.63)	0.510	1.11 (0.66-1.89)	0.687
	Grade IV	2.53 (1.82-3.52)	<0.001	1.56 (1.04-2.33)	0.031	1.18 (0.67-2.06)	0.567	0.75 (0.28-2.01)	0.569
Histology type	Adenocarcinoma	Ref		Ref		Ref		Ref	
	Mucinous adenocarcinoma	1.12 (0.94-1.34)	0.199	1.14 (0.93-1.40)	0.201	1.04 (0.78-1.39)	0.803	1.27 (0.85-1.89)	0.239
	Others	1.68 (1.26-2.24)	<0.001	1.41 (0.95-2.10)	0.087	0.70 (0.31-1.56)	0.379	0.79 (0.25-2.47)	0.687
Tumor size	< 5cm	Ref		Ref		Ref		Ref	
	≥5cm	1.40 (1.27-1.55)	<0.001	1.33 (1.18-1.49)	<0.001	1.13 (0.96-1.33)	0.133	1.13 (0.89-1.44)	0.316
Marital status	Married	Ref		Ref		Ref		Ref	
	Single	1.16 (0.97-1.39)	0.108	1.32 (1.09-1.61)	0.005	1.43 (1.09-1.88)	0.009	1.42 (0.95-2.14)	0.090
	Widowed	1.18 (1.06-1.32)	0.004	1.15 (1.01-1.31)	0.041	1.13 (0.94-1.36)	0.188	1.20 (0.91-1.58)	0.188
	Separated/Divorced	1.31 (1.09-1.57)	0.004	1.09 (0.86-1.36)	0.480	1.17 (0.86-1.59)	0.309	1.58 (1.04-2.40)	
CEA	Negative/Normal	Ref		Ref		Ref		Ref	
	Positive/Elevated	2.03 (1.83-2.24)	<0.001	1.87 (1.66-2.10)	<0.001	1.99 (1.69-2.34)	<0.001	1.59 (1.25-2.03)	<0.001
T stage	T1	Ref		Ref		Ref		Ref	
	T2	1.09 (0.77-1.56)	0.625	1.16 (0.78-1.72)	0.462	1.33 (0.80-2.23)	0.277	1.43 (0.77-2.68)	0.261
	T3	3.11 (2.30-4.21)	<0.001	3.23 (3.00-4.53)	<0.001	3.13 (2.00-4.90)	<0.001	2.18 (1.24-3.82)	0.007
	T4	8.08 (5.92-11.03)	<0.001	7.80 (5.49-11.09)	<0.001	5.89 (3.66-9.49)	<0.001	3.53 (1.87-6.66)	<0.001
N stage	N0	Ref		Ref		Ref		Ref	
	N1	2.54 (2.26-2.86)	<0.001	2.73 (2.39-3.12)	<0.001	2.10 (1.75-2.53)	<0.001	1.55 (1.19-2.03)	0.001
	N2	5.12 (4.51-5.81)	<0.001	5.23 (4.51-6.06)	<0.001	3.51 (2.83-4.36)	<0.001	1.63 (1.10-2.42)	0.015
Radiation	None/Unknown	Ref		Ref				Ref	
	Yes	1.44 (0.86-2.39)	0.163	2.27 (1.44-3.57)	<0.001			0.60 (0.08-4.27)	0.609
Chemotherapy	None/Unknown	Ref		Ref		Ref		Ref	
	Yes	1.56 (1.40-1.74)	<0.001	1.74 (1.54-1.97)	<0.001	1.76 (1.48-2.08)	<0.001	1.32 (1.02-1.72)	0.035

HR Hazard ratio



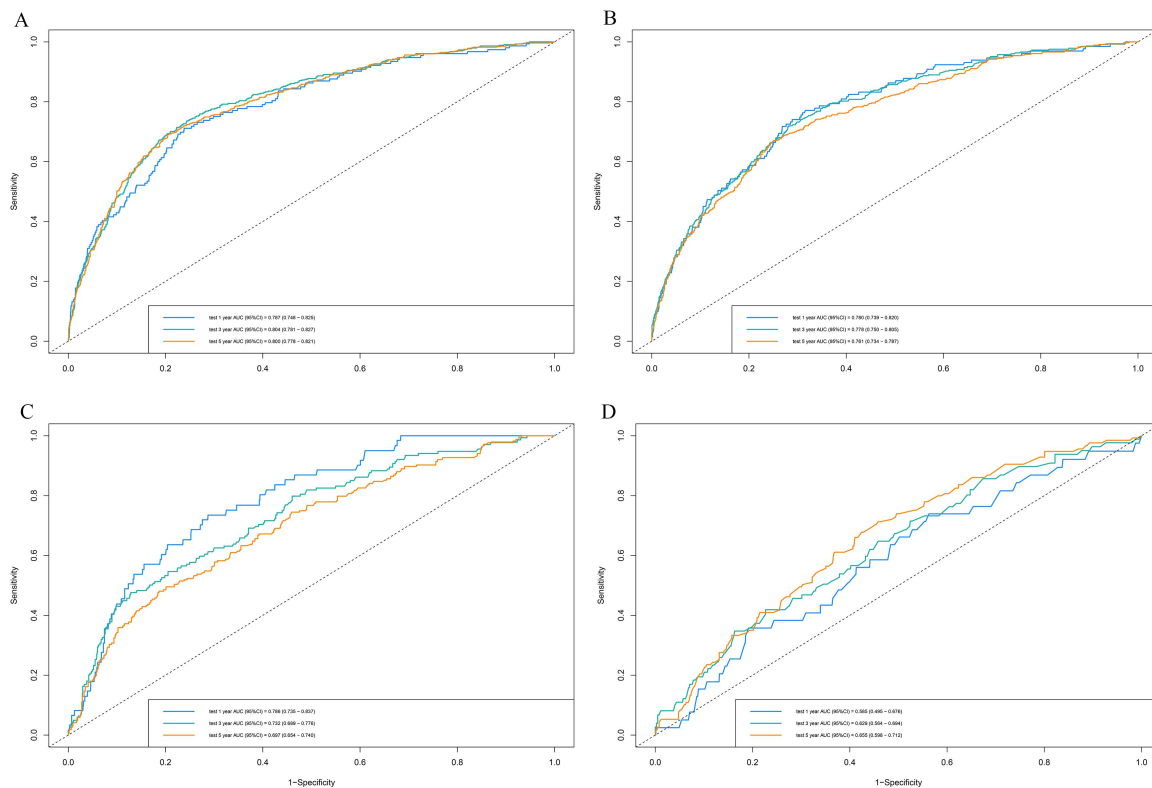
Supplementary Figure 1

Flow diagram for enrollment of the study cohort.



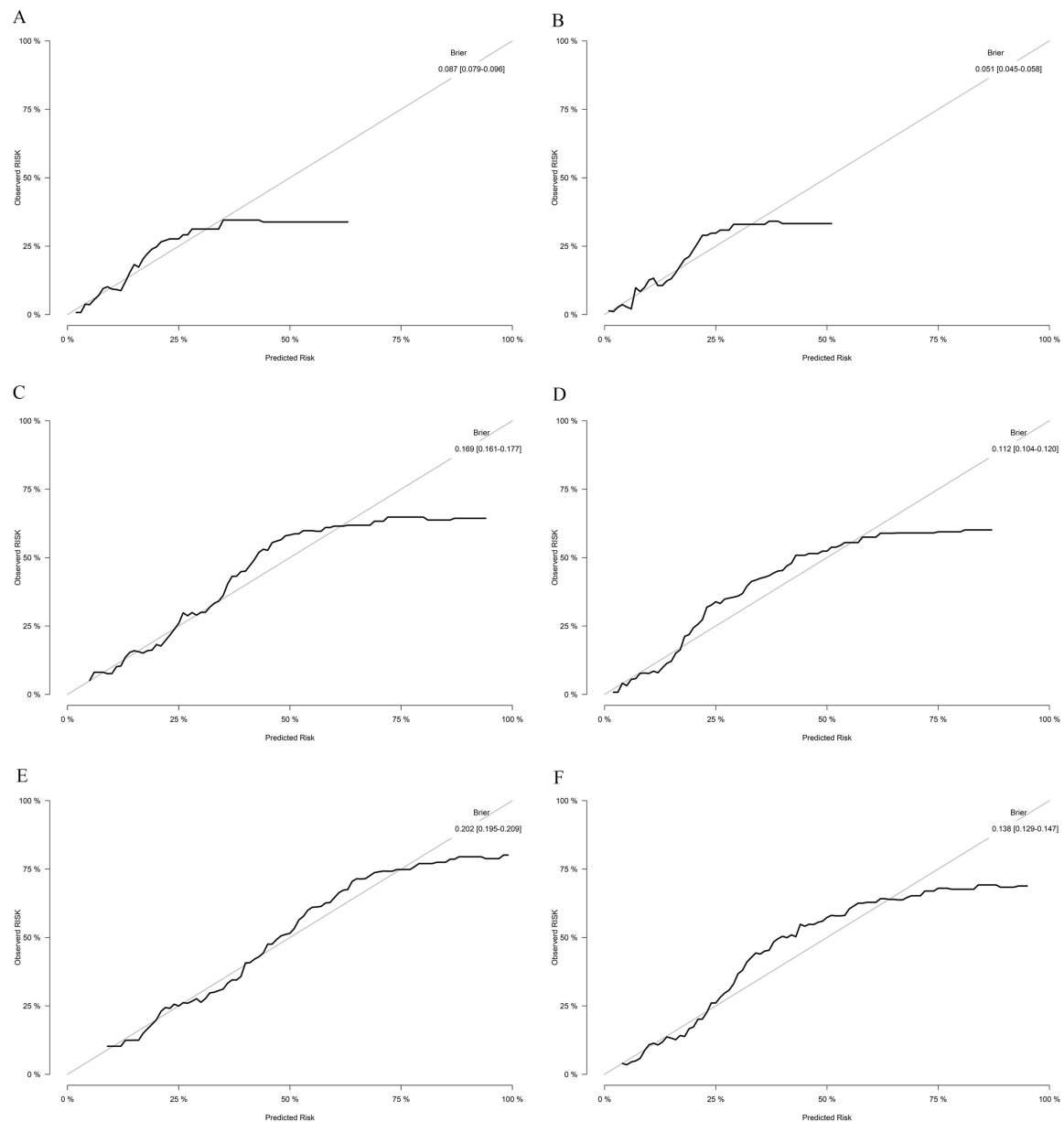
Supplementary Figure 2

ROC curve analyses in the test cohort for one-, three-, and five-year overall survival in baseline (A), one-year (B), three-year (C), and five-year (D) survivors. ROC, receiver operator characteristic curve; AUC, area under the curve.



Supplementary Figure 3

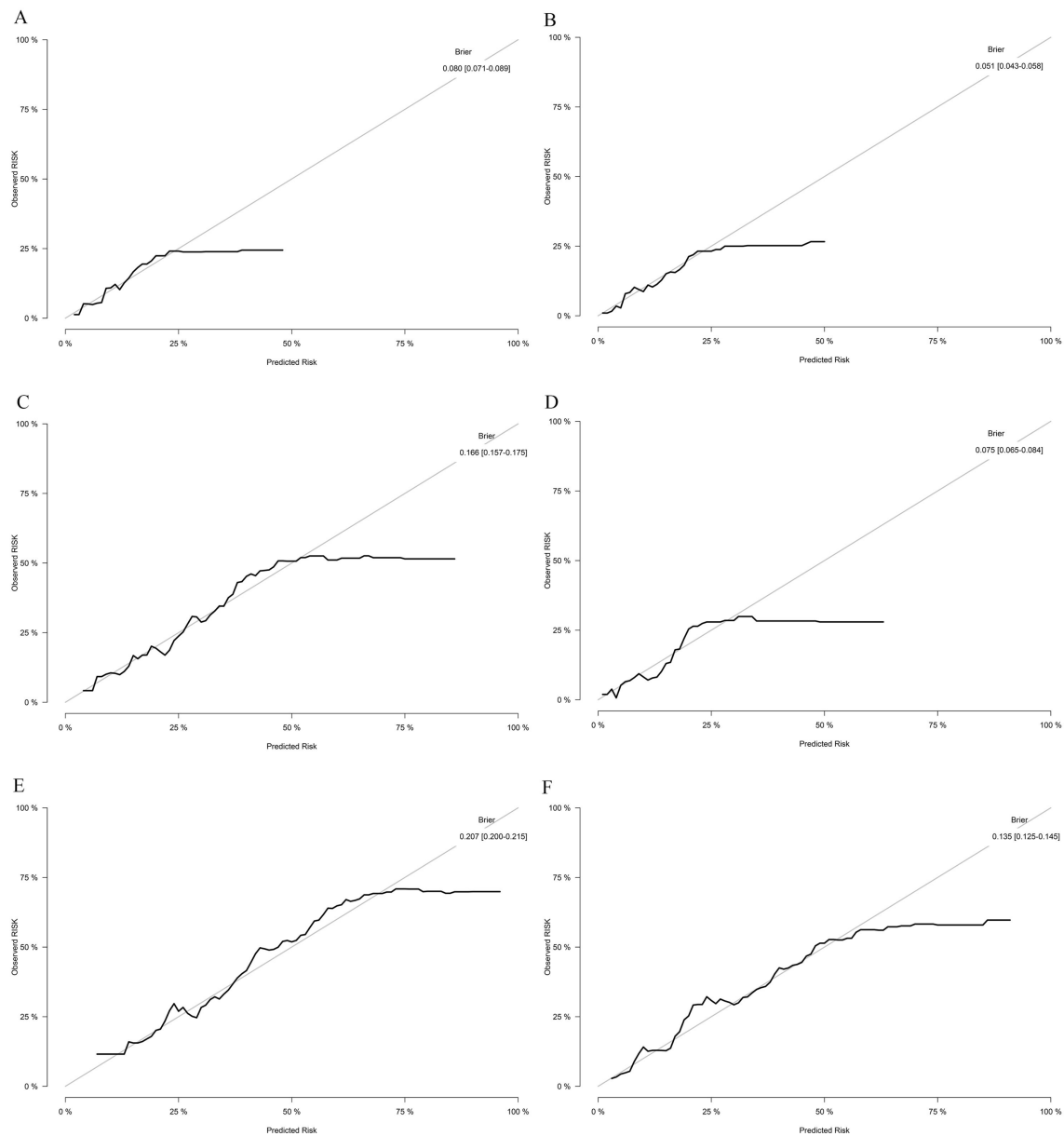
ROC curve analyses in the test cohort for one-, three-, and five-year colon-specific survival in baseline (A), one-year (B), three-year (C), and five-year (D) survivors. ROC, receiver operator characteristic curve; AUC, area under the curve.



Supplementary Figure 4

Calibration curves of baseline nomograms.

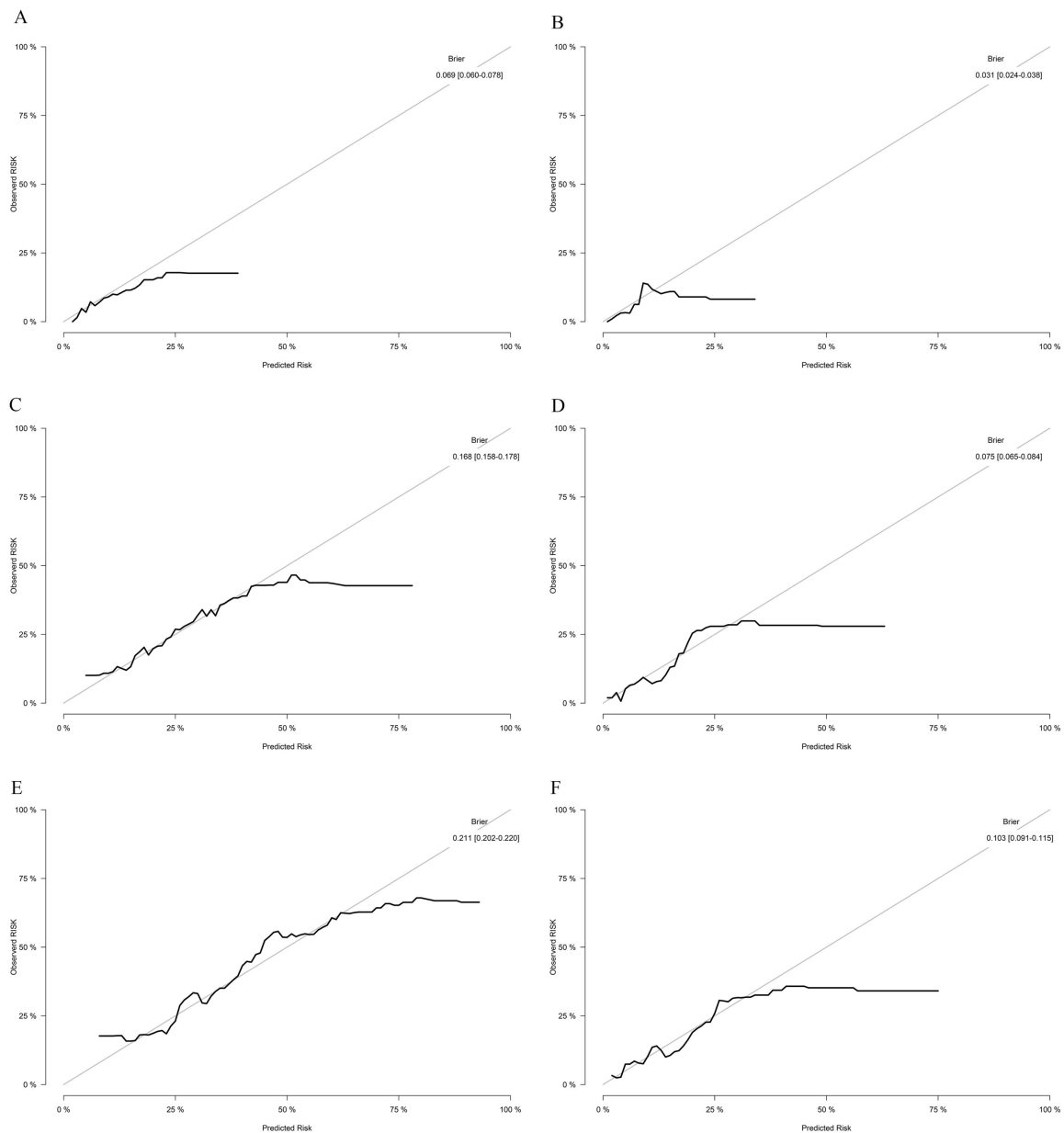
(A, C and E) showed Calibration curves of nomograms in predicting one-, three-, and five-year OS, respectively. (B, D and F) showed Calibration curves of nomograms in predicting one-, three-, and five-year CSS, respectively. OS, overall survival; CSS, colon-specific survival.



Supplementary Figure 5

Calibration curves of one-year survivors nomograms.

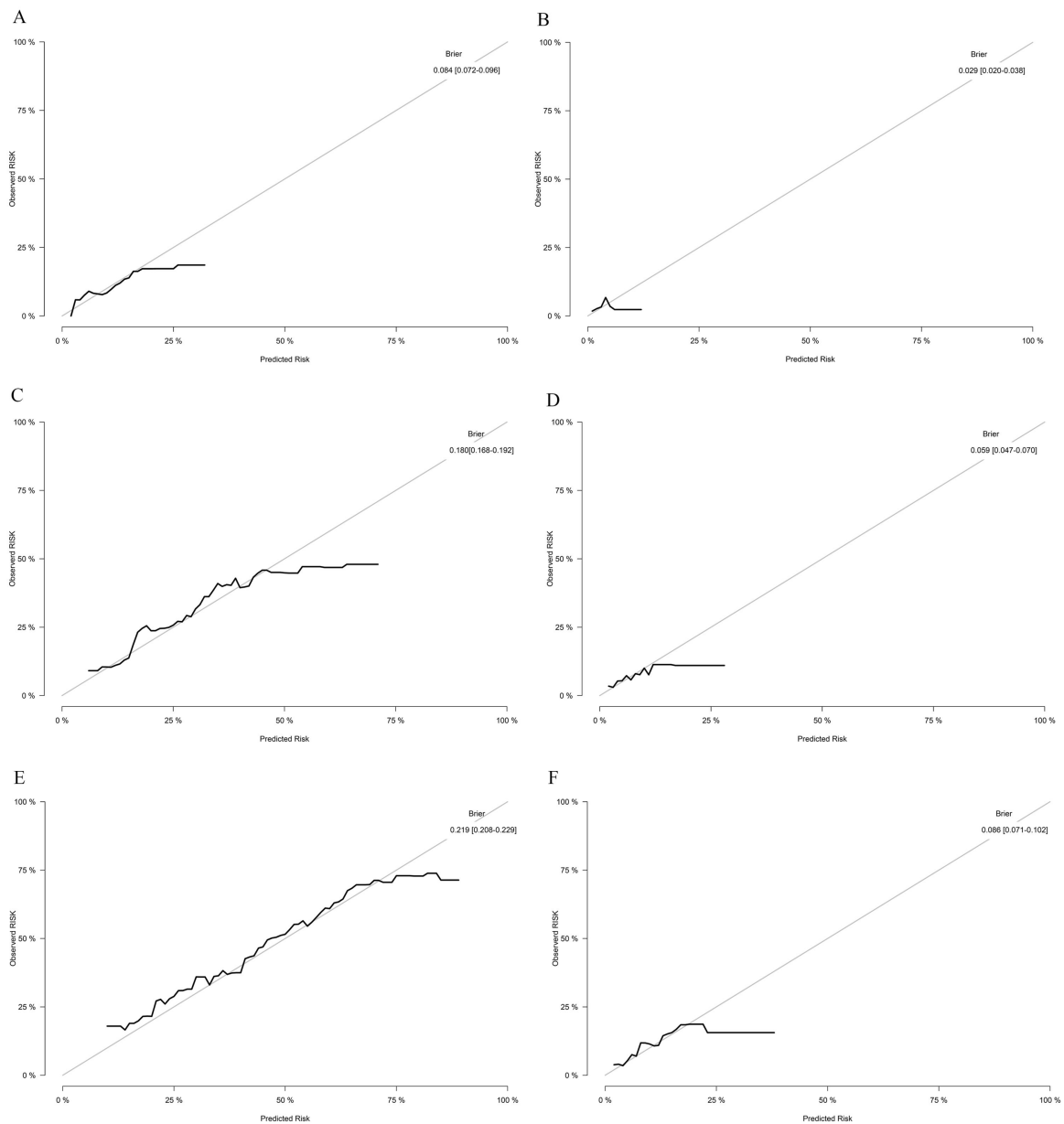
(A, C and E) showed Calibration curves of nomograms in predicting one-, three-, and five-year OS, respectively. (B, D and F) showed Calibration curves of nomograms in predicting one-, three-, and five-year CSS, respectively. OS, overall survival; CSS, colon-specific survival.



Supplementary Figure 6

Calibration curves of three-year survivors nomograms in the training cohort.

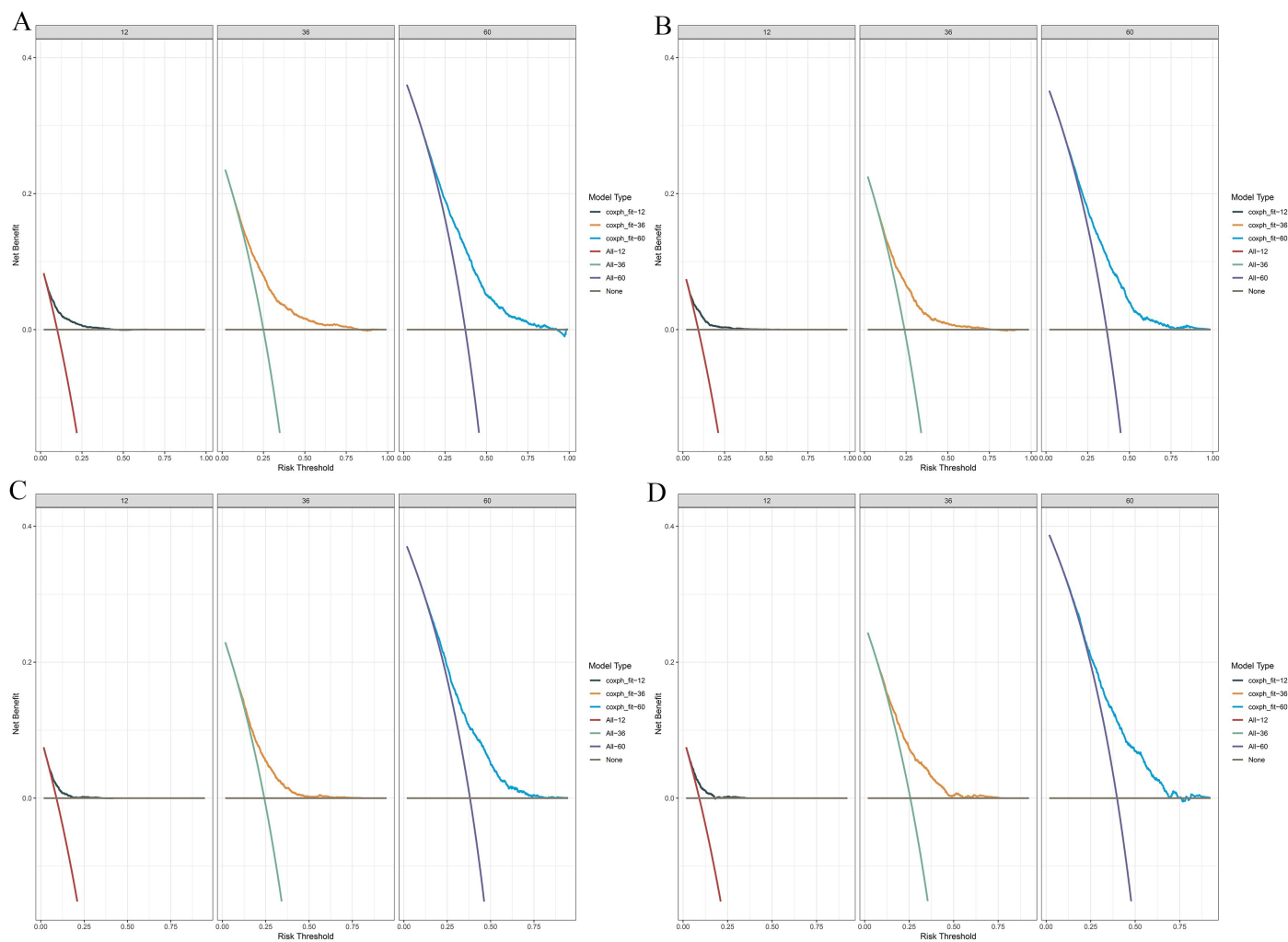
(A, C and E) showed Calibration curves of nomograms in predicting one-, three-, and five-year OS, respectively. (B, D and F) showed Calibration curves of nomograms to predict one-, three-, and five-year CSS, respectively. OS, overall survival; CSS, colon-specific survival.



Supplementary Figure 7

Calibration curves of five-year survivors nomograms in the training cohort.

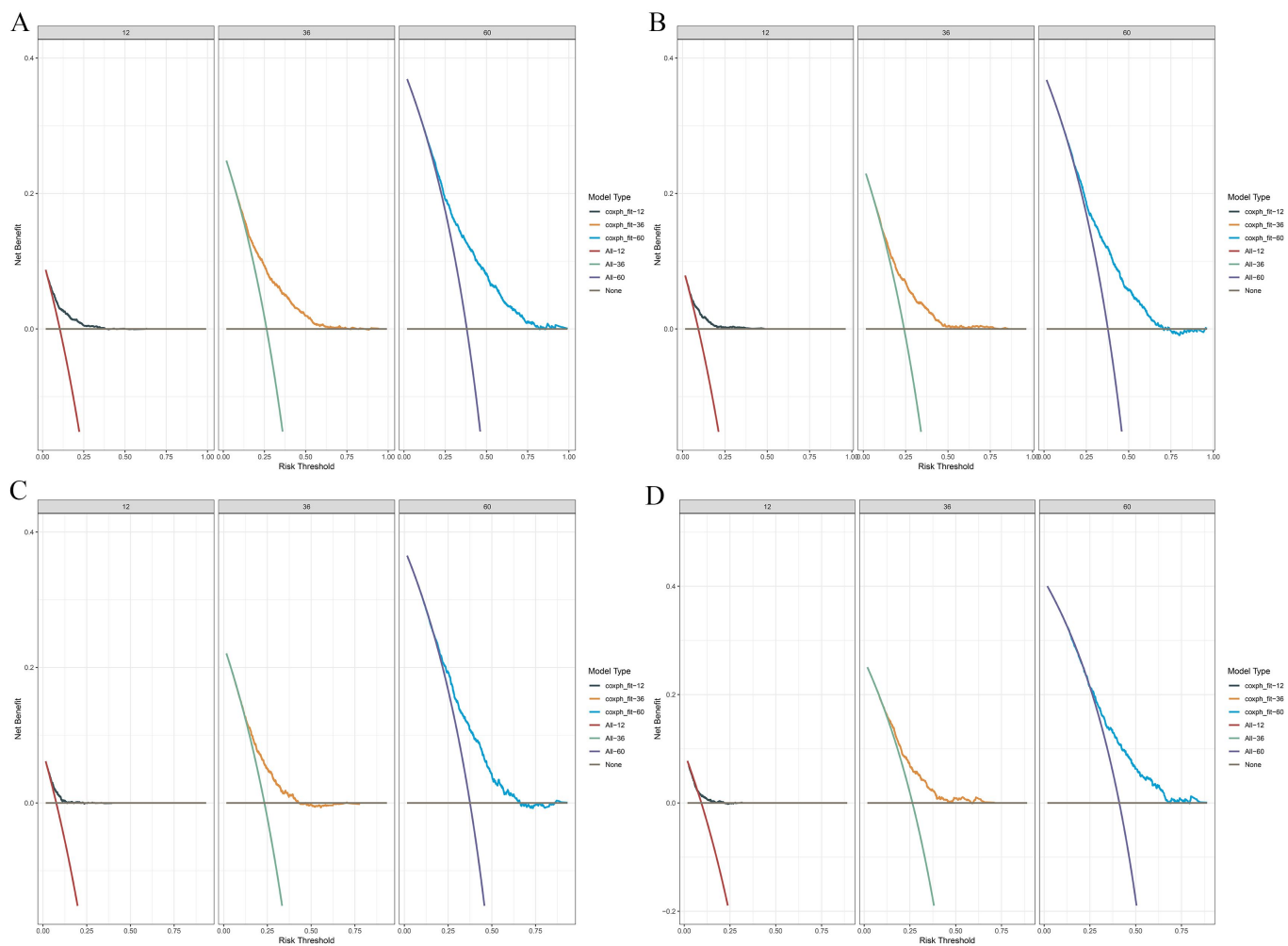
(A, C and E) showed Calibration curves of nomograms in predicting one-, three-, and five-year OS, respectively. (B, D and F) showed Calibration curves of nomograms in predicting one-, three-, and five-year CSS, respectively. OS, overall survival; CSS, colon-specific survival.



Supplementary Figure 8

Decision curve analysis for overall survival nomograms in the training cohort.

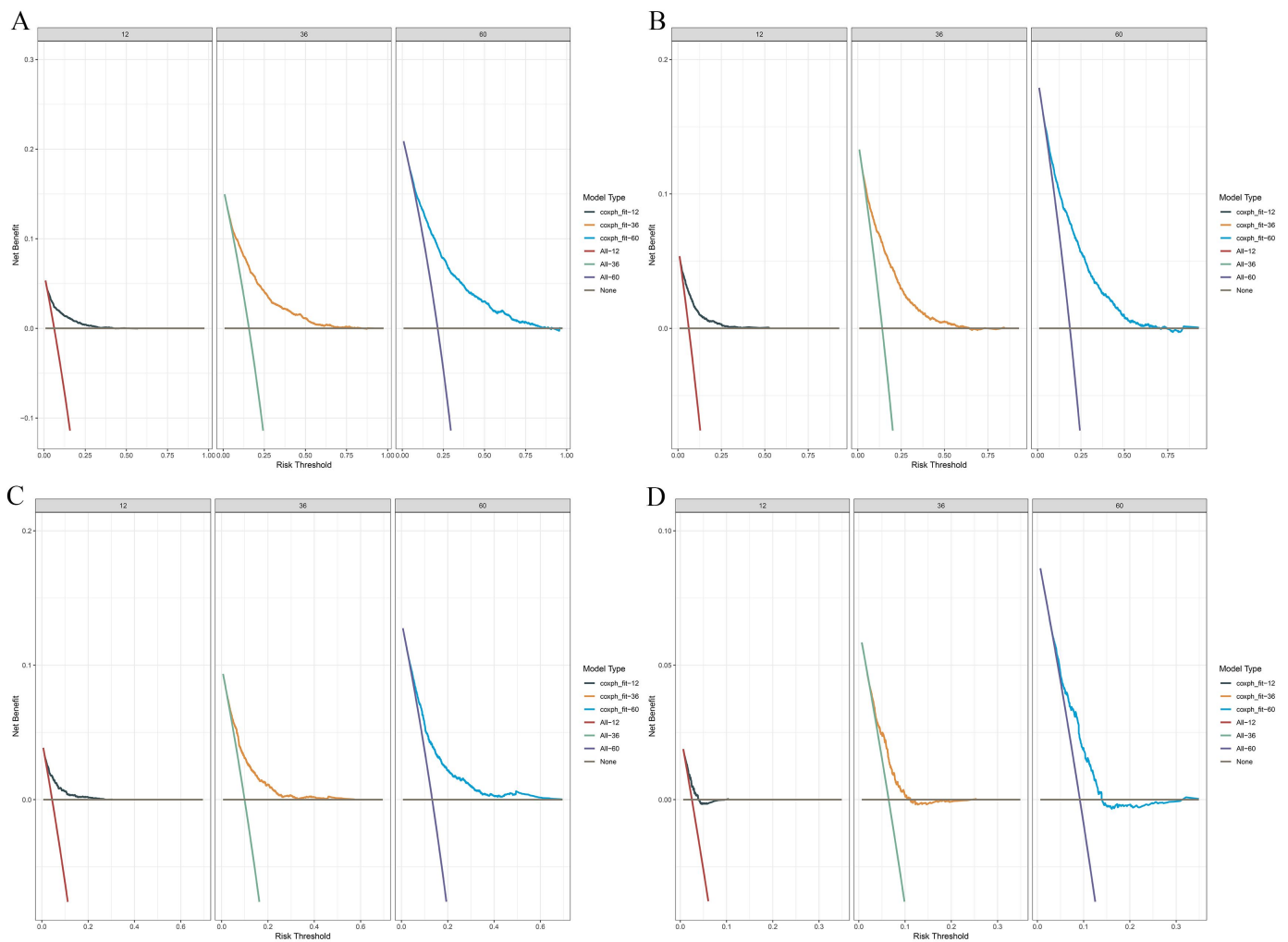
A, patients at baseline; B, Patients who have survived for one year; C, Patients who have survived for three years; D, Patients who have survived for five years.



Supplementary Figure 9

Decision curve analysis for overall survival nomograms in the test cohort.

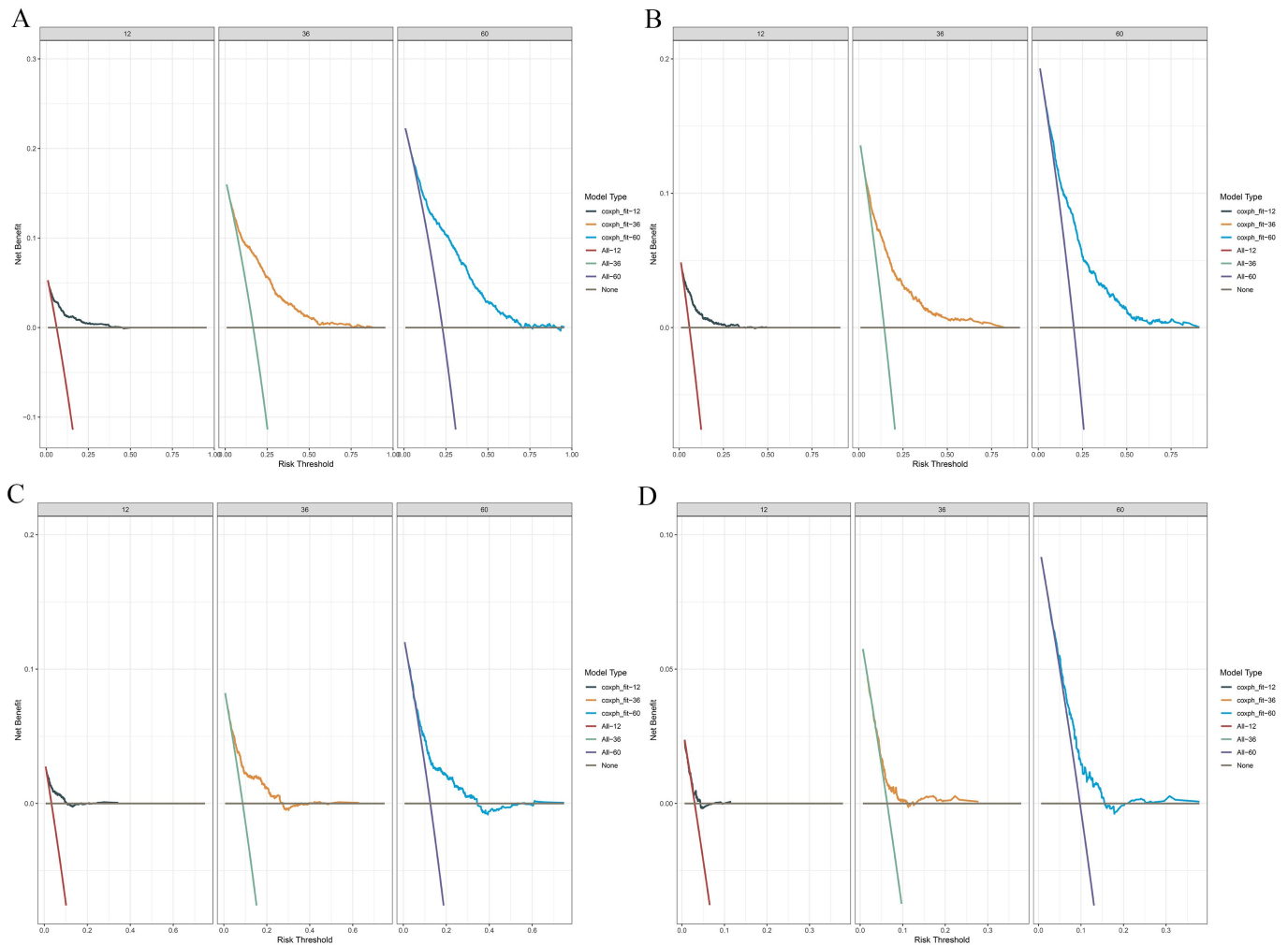
A, patients at baseline; B, patients who have survived for one year; C, patients who have survived for three years; D, patients who have survived for five years.



Supplementary Figure 10

Decision curve analysis for colon-specific survival nomograms in the training cohort.

A, patients at baseline; B, patients who have survived for one year; C, patients who have survived for three years; D, patients who have survived for five years.



Supplementary Figure 11

Decision curve analysis for colon-specific survival nomograms in the test cohort.

A, patients at baseline; B, patients who have survived for one year; C, patients who have survived for three years; D, patients who have survived for five years.

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Reviewer #1 (Public Review):

Summary:

This study assessed conditional survival in elderly patients with non-metastatic colon cancer who underwent colectomy. The study found that 5-year conditional overall survival rates exhibited a slight increase initially, followed by a decrease over time. In contrast, 5-year conditional colon-specific survival rates consistently improved over the same period.

Nomograms were developed to predict survival probabilities at baseline and for patients surviving 1, 3, and 5 years post-diagnosis, with good predictive performance. The study concludes that conditional survival offers valuable insights into medium- and long-term survival probabilities for these patients.

Strengths:

The strengths of this study include robust study design, methodology, statistical analysis, and interpretation of the findings. Utilizing a well-known database for the analysis is another strength. Differentiating overall survival and colon-specific survival rates could be another one. Focusing on elderly patients with this condition is another major point. Providing nomograms for an easier implication of the findings in real-world clinical practice is a major strength of the study.

Weaknesses:

Relying on only one database of patients and narrowing down the population to only elderly patients who underwent colectomy could be mentioned as a weakness. Less generalizability of the findings for other populations and not including more diverse databases is a major weakness of this study. The good predictive capabilities of the developed tools are another weakness that could be improved to be excellent.

<https://doi.org/10.7554/eLife.97167.1.sa2>

Reviewer #2 (Public Review):

Summary:

The authors assessed the conditional survival of elderly patients with non-metastatic colon cancer who had survived a certain length of time after colectomy. They used data from the Surveillance, Epidemiology, and End Results (SEER) registry to conduct a conditional survival analysis providing estimates of conditional survival rates as well as an analysis of which variables were most important for survival at baseline, one year, three years, and five years.

Strengths:

- The authors used SEER data, providing them with long-term follow-up, and thoroughly considered a wide range of variables related to cancer mortality.
- The authors did a thorough job of assessing the predictive ability of their models.
- The authors used conditional survival, providing estimates of survival that are meaningful for patients/physicians, making them useful for clinical practice.

Weaknesses:

- The paper would have benefited from a more thorough explanation of why the methods were improvements on existing approaches.
- This study was primarily interested in cancer mortality, and compared it to the secondary outcome of death from any cause. The study would have benefited from modeling death from non-cancer causes (the competing risk) in addition to death from colon cancer, rather than comparing only to the composite endpoint of death from any cause.
- When considering a cause-specific hazard, as done with cancer survival in this paper, it would be better to consider the cumulative incidence function rather than Kaplan Meier, since it does not assume the independence of the events like Kaplan Meier does. For this reason, the paper would benefit from focusing on the results of the adjusted cause-specific hazard models (rather than the unadjusted conditional survival estimates done using Kaplan

Meier estimates shown in Figure 1 and conducting a parallel analysis for death from other causes.

- The authors mention that they consider disparities using a log-rank test. For the same reason as above, is not the best approach when dealing with competing risks as it depends on Kaplan Meier curves. The log-rank test may be fine if there is no strong dependence between the two causes of death, but the paper would benefit from some discussion of that choice, or sensitivity analysis by comparison to other approaches.

- The variables for the adjusted models were chosen with univariate Cox regression analysis, with any variables having a p-value less than 0.05 being included in the adjusted. Another approach, which may have made the models more easily comparable, would be to choose the variables that are relevant based on prior literature and include them in the multivariate model regardless of significance. The paper would benefit from a discussion of what is gained by excluding some variables from some models.

<https://doi.org/10.7554/eLife.97167.1.sa1>

Reviewer #3 (Public Review):

Summary:

This article uses a subset of data from the SEER cancer registry to develop nomograms, a patient-facing risk prediction tool, for predicting overall and cancer-specific survival in elderly patients who underwent colectomy for the treatment of non-metastatic colon cancer. A unique contribution is the intent to provide conditional predictions, i.e. given that you have survived for x years from your diagnosis, what is your probability of survival for an additional y years? Although the goal is a useful one, the approach is unfortunately hampered by some important weaknesses.

Strengths:

Predicting conditional overall survival is a useful, patient-oriented goal.

The data source is the high-quality SEER cancer registry.

Weaknesses:

Using Kaplan-Meier methodology to estimate the survival distribution for a time-to-event in the presence of another competing time-to-event (in this case: estimating colon-specific survival in the presence of death from other causes) will generally over-estimate the event rate. The reported colon-specific survival probabilities are probably biased downwards from their true values. See <https://pubmed.ncbi.nlm.nih.gov/10204198/>

A similar concern would apply to the use of the cause-specific Cox model, and thus also the nomogram, to predict absolute (conditional) survival.

The p-value-based methodology for determining which predictors should be included in the nomogram is rudimentary. More modern variable selection methods, e.g. the Lasso, would have been preferred.

Related to the above comment, some predictors are present for the conditional survival nomogram for time t, then absent for time t+1, then present again for time t+2. A cancer site is an example of such a predictor. From a face validity perspective, this doesn't really make sense. Ideally, predictors would not enter, then leave, and then re-enter a model.

Many observations were excluded due to missingness in predictors, e.g. >10000 were excluded due to unknown CEA (Supplementary Figure 1). Given the number of observations dropped due to missingness in the predictors, ideally an attempt would have been made to incorporate the partial information available in these data.

Details are lacking on how the AUCs and Brier scores were calculated in the presence of censoring / competing events, which limits the reader's understanding of the results.

It is not clear why a nomogram would be preferred to an online risk prediction calculator.

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